Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

Outline

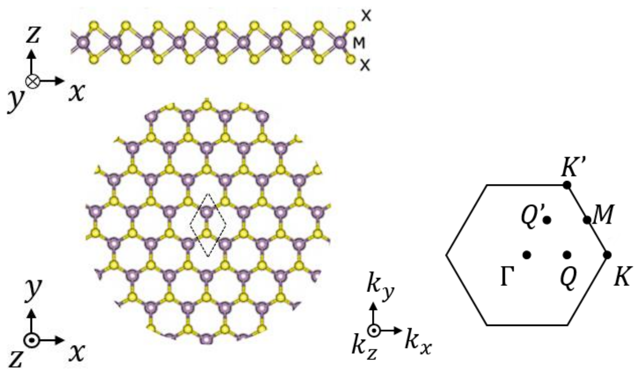
Outline

1. Introduction

What material ?

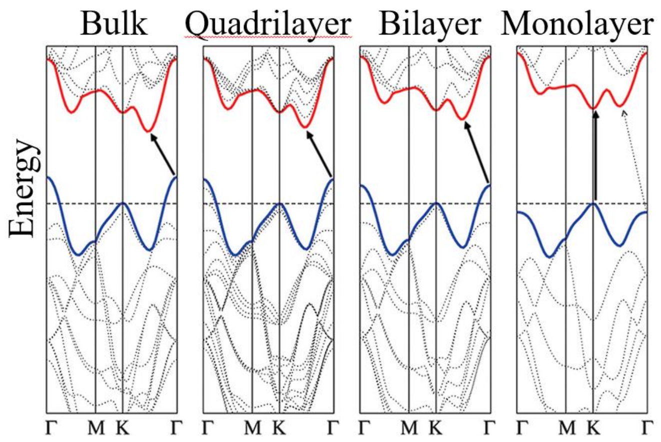
1. Introduction

- Two dimensional (2D) Transition metal dichalcogenide (TMD)



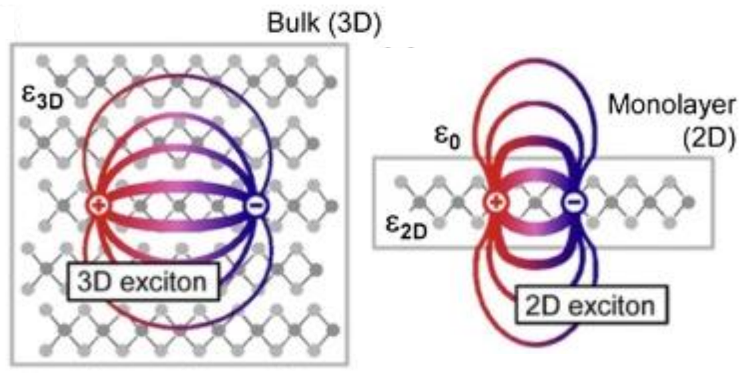
- Structures of a typicle 2D TMD material

A. Krasnok & A. Alù, *Appl. Sci.* **8**, 1157 (2018)



- MoS_2, calculated band structure

A. Splendiani *et al.*, *Nano Lett.* **10**, 1271 (2010)



- Screening in materials

M. Velický & P. S. Toth, *Appl. Mater. Today* **8**, 68 (2017)

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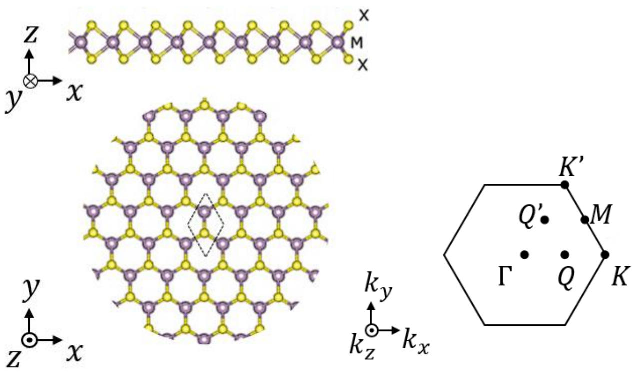

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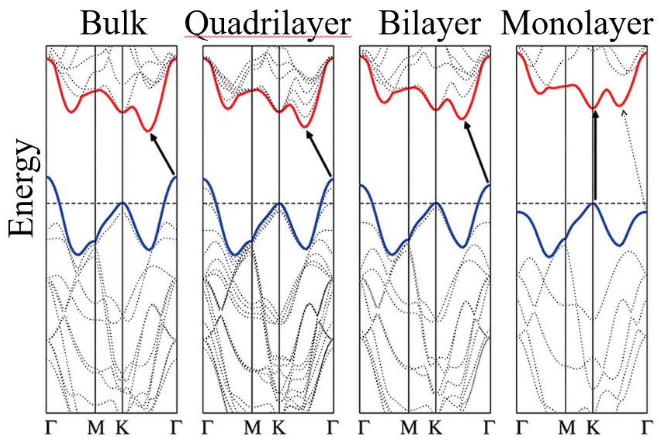

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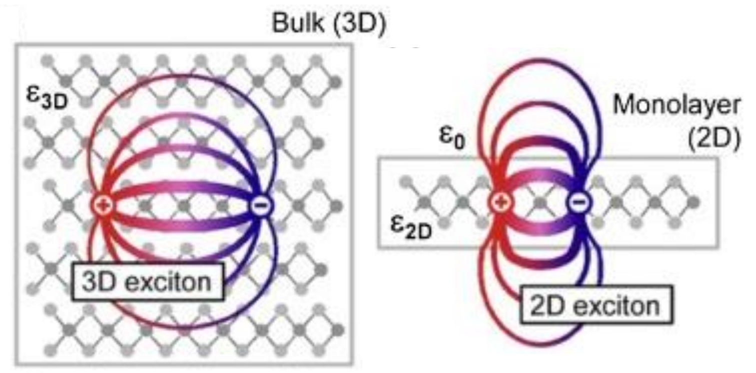

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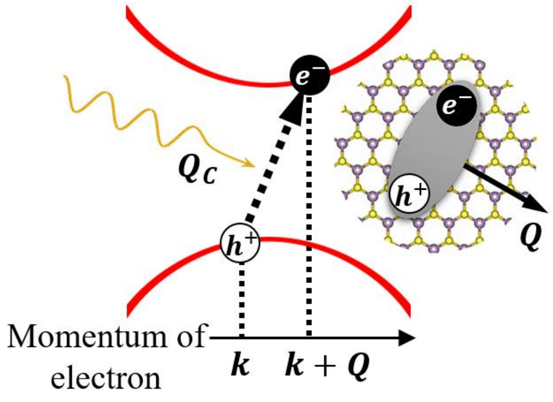

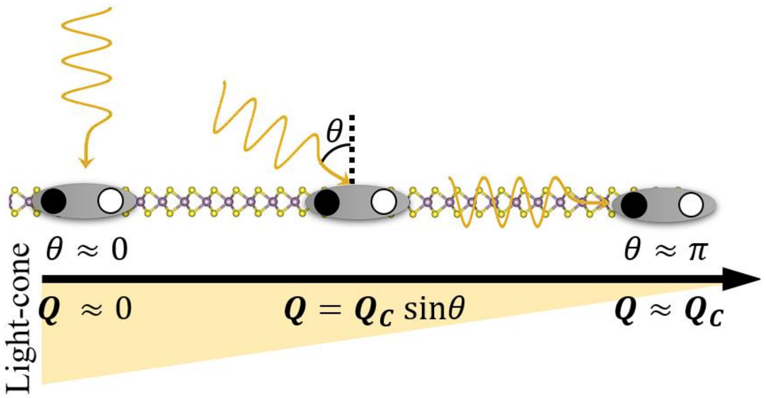

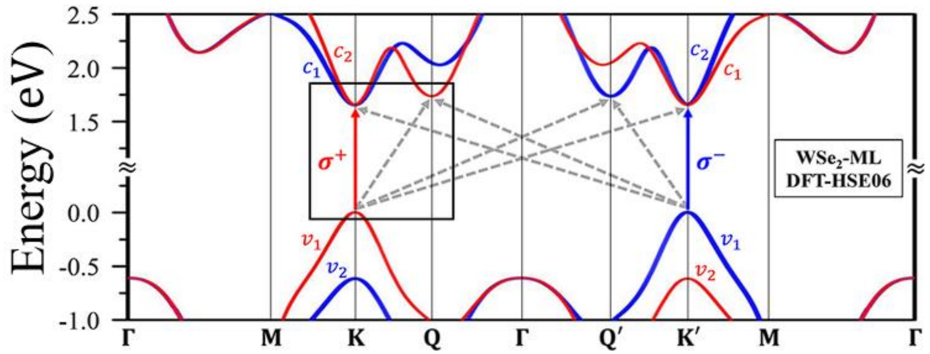

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G.-H. Peng *et al.*, *Nano Lett.* **19**, 2299 (2019)

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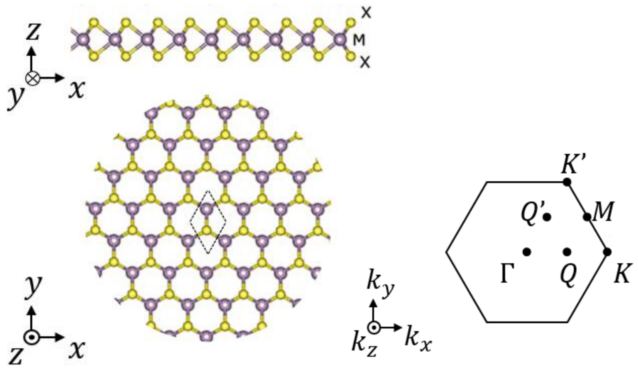

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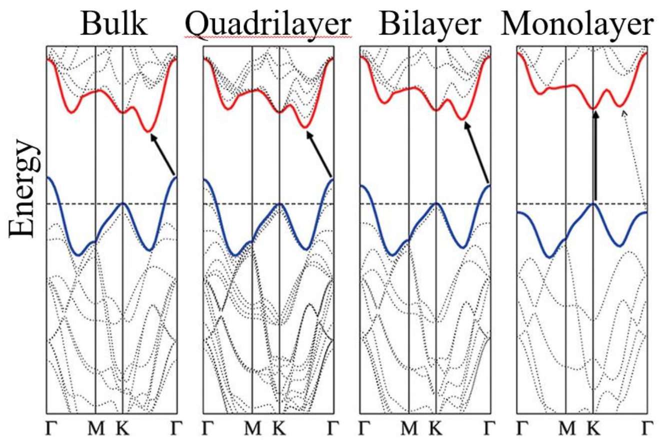

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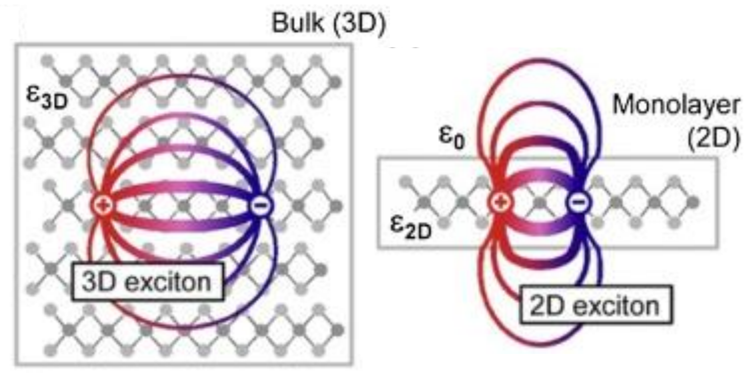

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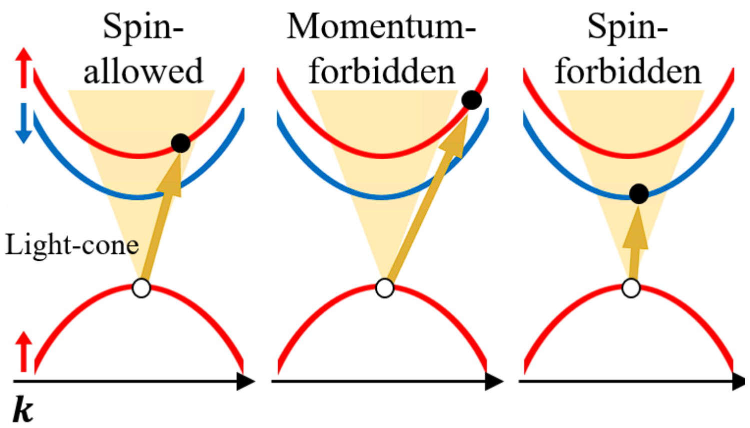

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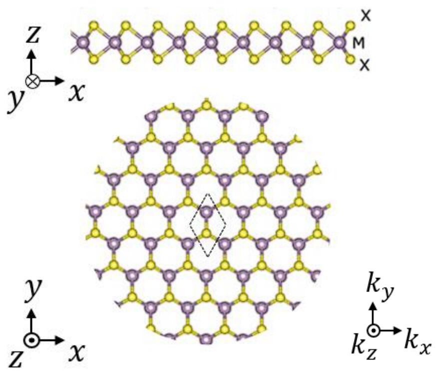

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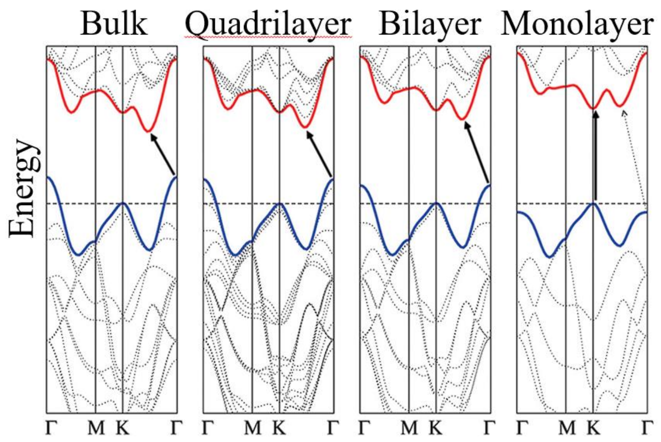

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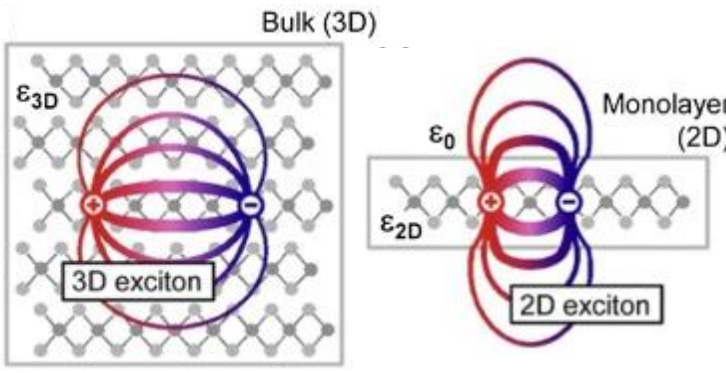

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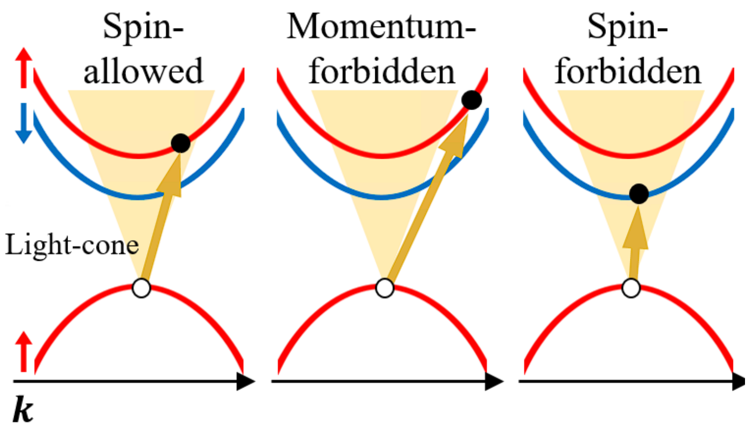

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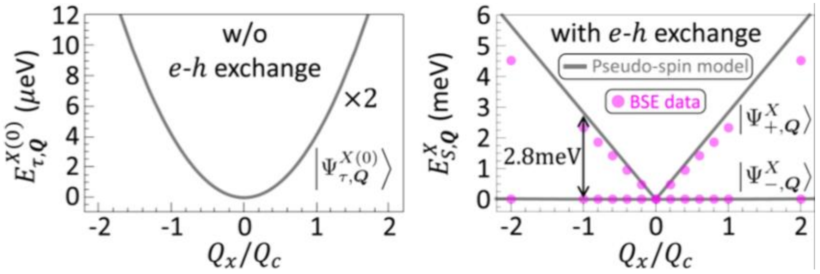

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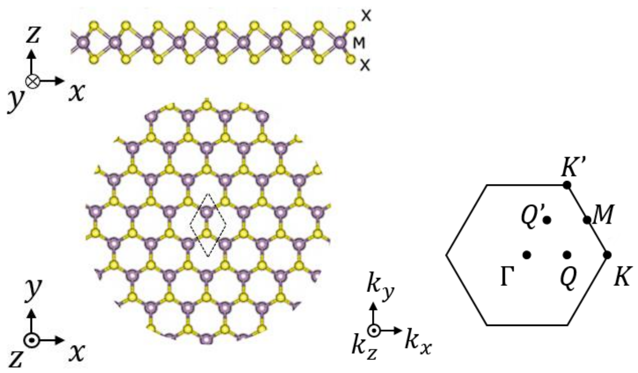

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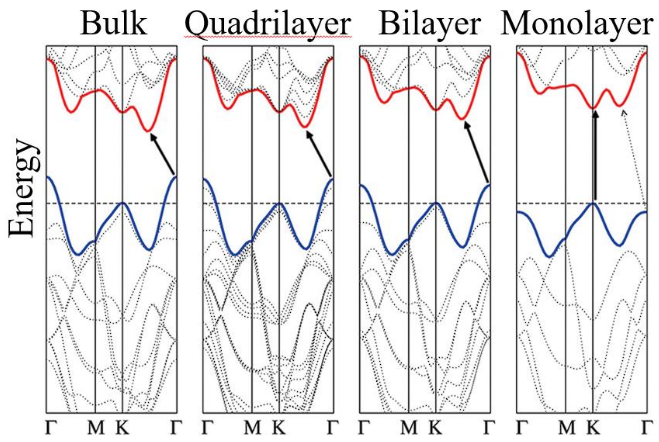

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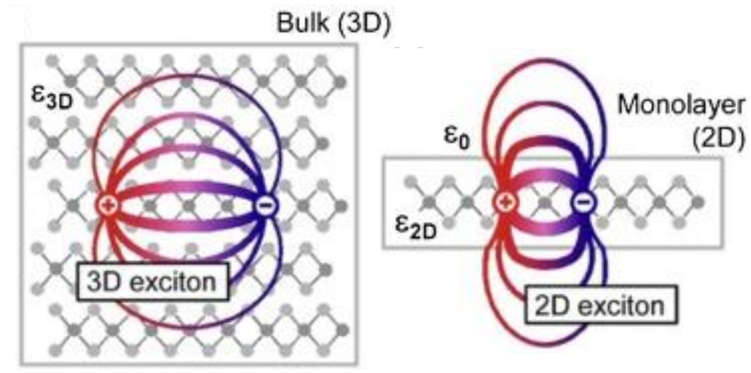

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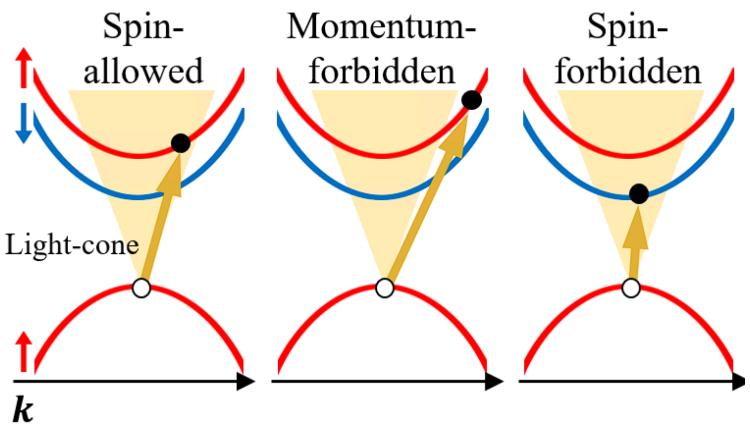

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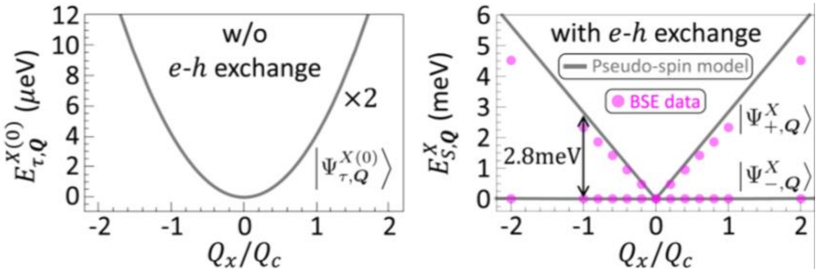

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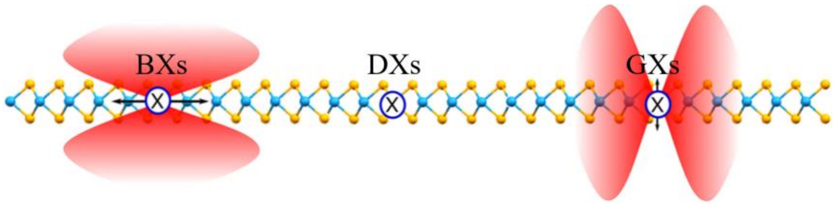


- Excitons categorized by radiation patterns

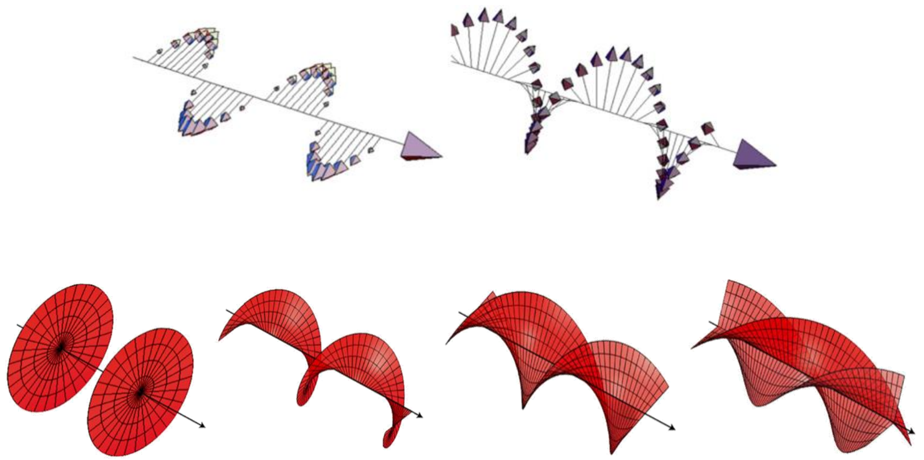
L. M. Schneider *et al.*, *Sci. Rep.* **10**, 8091 (2020)

1. Introduction

What light information ?

1. Introduction

- Angular momentum in light



- Polarization of light / Helical phase fronts for $\ell = 0, 1, 2, 3$

A. M. Yao & M. J. Padgett, *Adv. Opt. Photonics* **3**, 161 (2011)

- Spin angular momentum (SAM)
  - > circular polarizations
  - > $\sigma\hbar$ per photon ($\sigma = \pm 1$)

- Orbital angular momentum (OAM)
  - > helical wave front (ex. Laguerre-Gaussian (LG) beam)
  - > $\ell\hbar$ per photon ($\ell = 0, \pm 1, \pm 2, \pm 3 \dots \dots \pm Z$)

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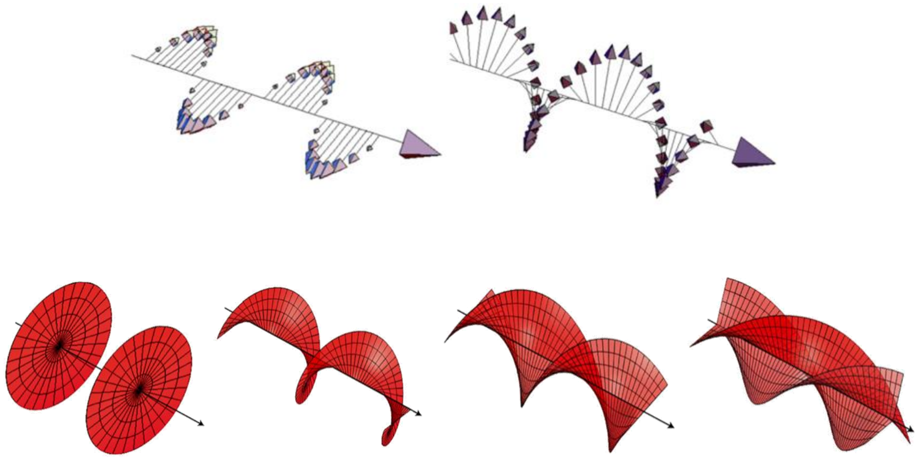

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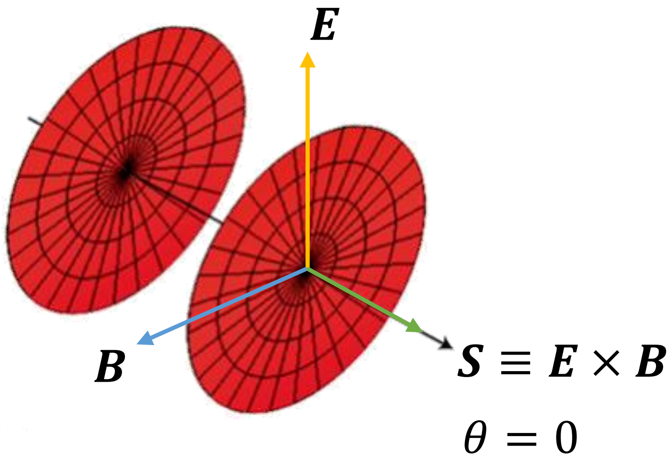

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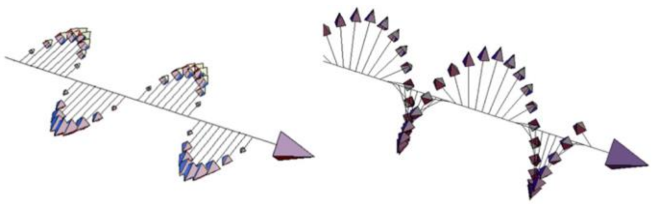

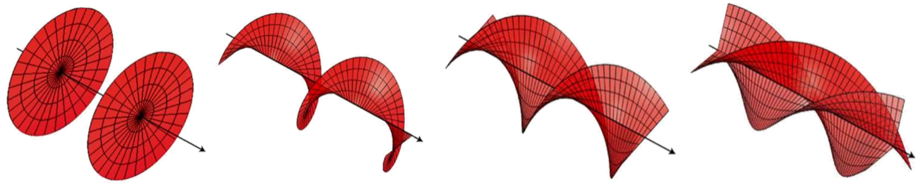

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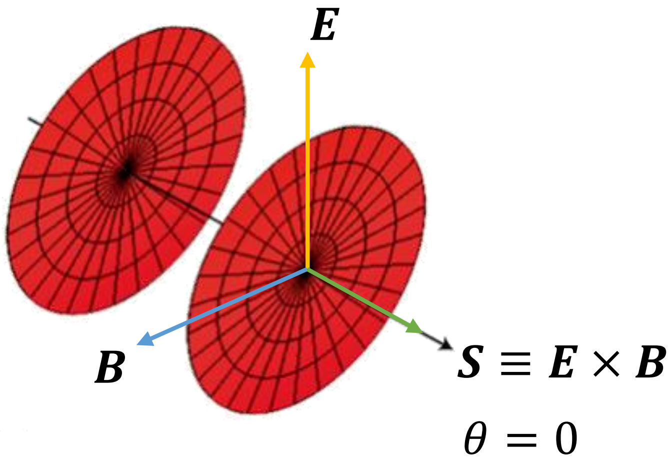

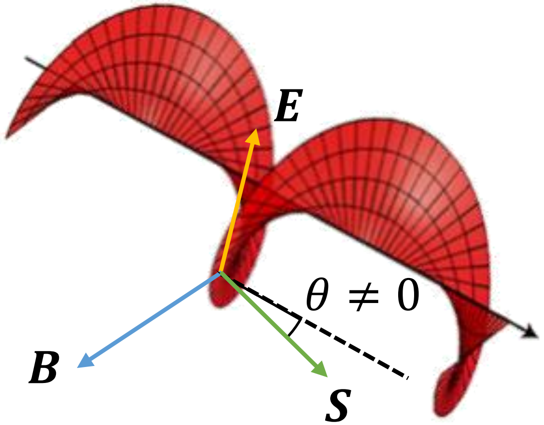

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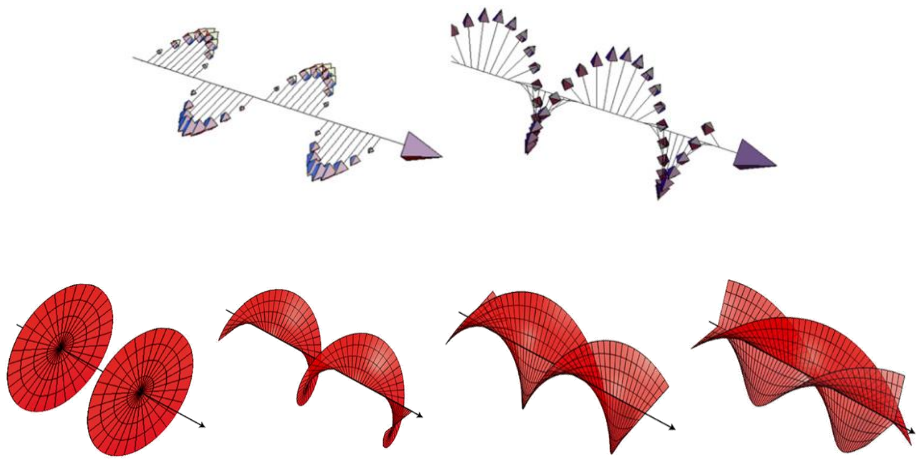

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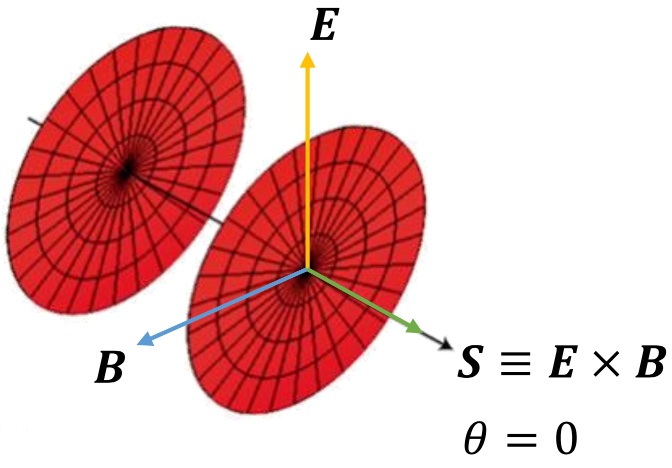

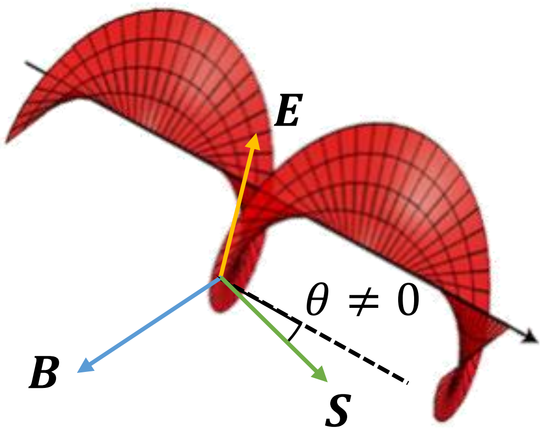

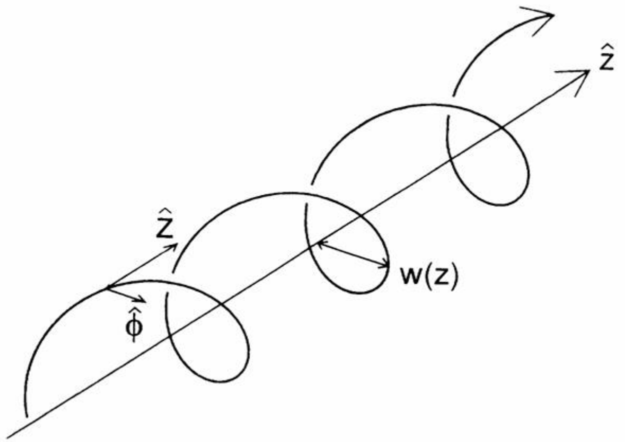

- Spiraling Poynting vector
L. Allen *et al.*, *Phys. Rev. A* **45**, 8185 (1992)

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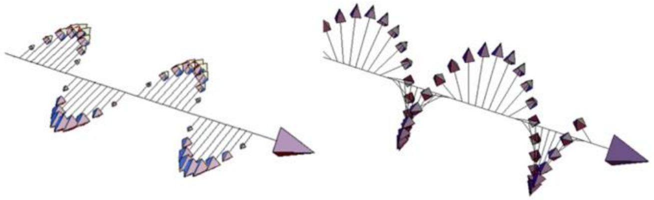

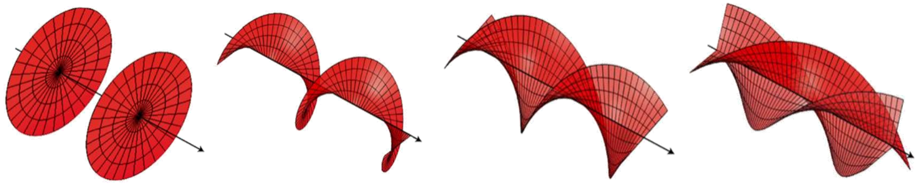

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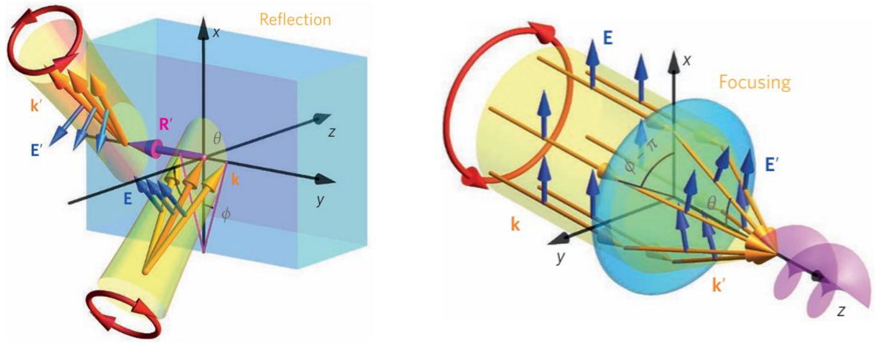

- Some cases for SOI

K. Y. Bliokh *et al.*, *Nat. Photonics* **9**, 796 (2015).

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- Spin & Orbital angular momentum interaction (SOI)
  - > circularly polarized laser beam reflected or refracted by medium inhomogeneity
  - > Tight focusing of a paraxial beam
  - > paraxial beam propagates in optical fibres
  - > other ……

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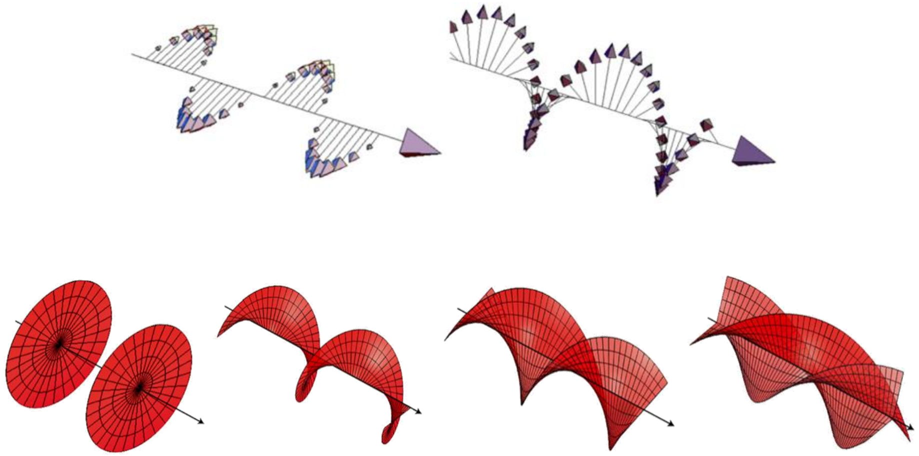

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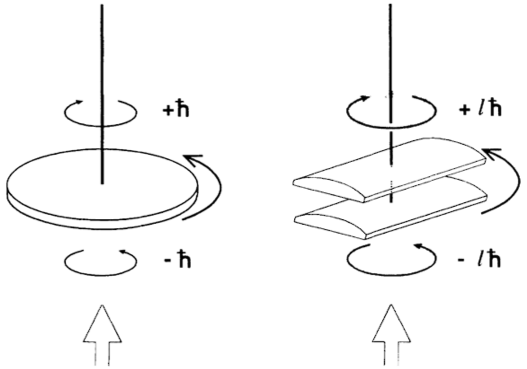
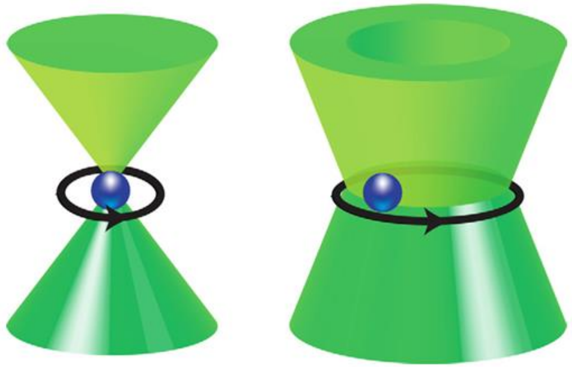
- Light matter interaction



- Measurements of SAM and OAM

R. A. Beth, *Phys. Rev.* **50**, 115 (1936)/L. Allen *et al.*, *Phys. Rev. A* **45**, 8185 (1992)



- Transfer of SAM and OAM in optical tweezers

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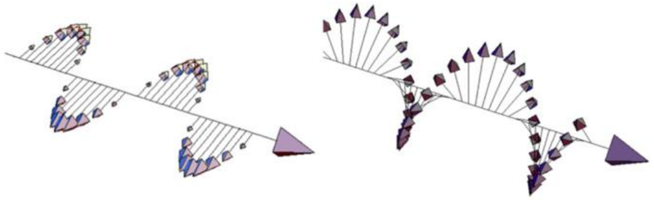

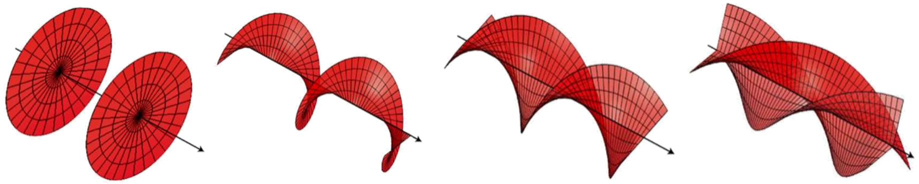

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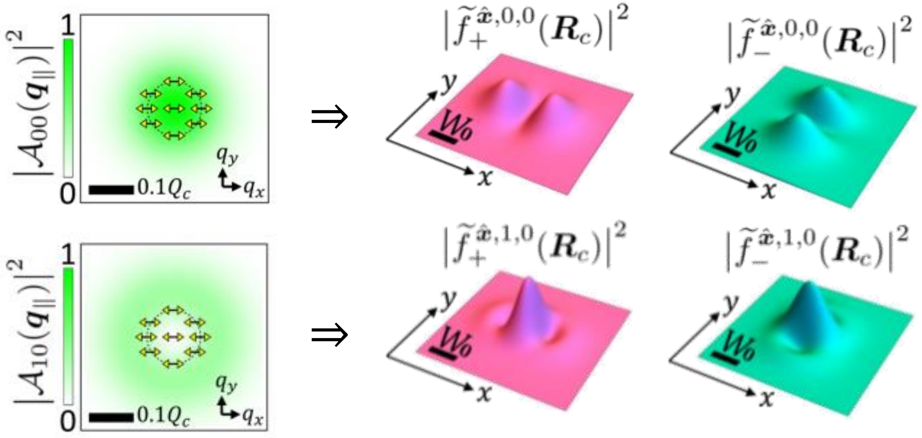

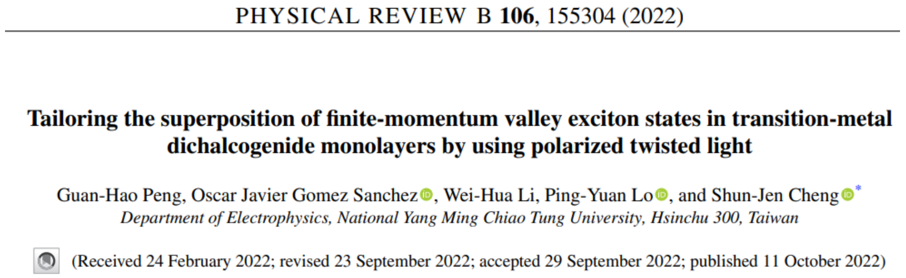
PHYSICAL REVIEW B **106**, 155304 (2022)

Tailoring the superposition of finite-momentum valley exciton states in transition-metal dichalcogenide monolayers by using polarized twisted light

Guan-Hao Peng, Oscar Javier Gomez Sanchez, Wei-Hua Li, Ping-Yuan Lo, and Shun-Jen Cheng*
Department of Electrophysics, National Yang Ming Chiao Tung University, Hsinchu 300, Taiwan

(Received 24 February 2022; revised 23 September 2022; accepted 29 September 2022; published 11 October 2022)

- OAM can influence exciton's wave packet structures

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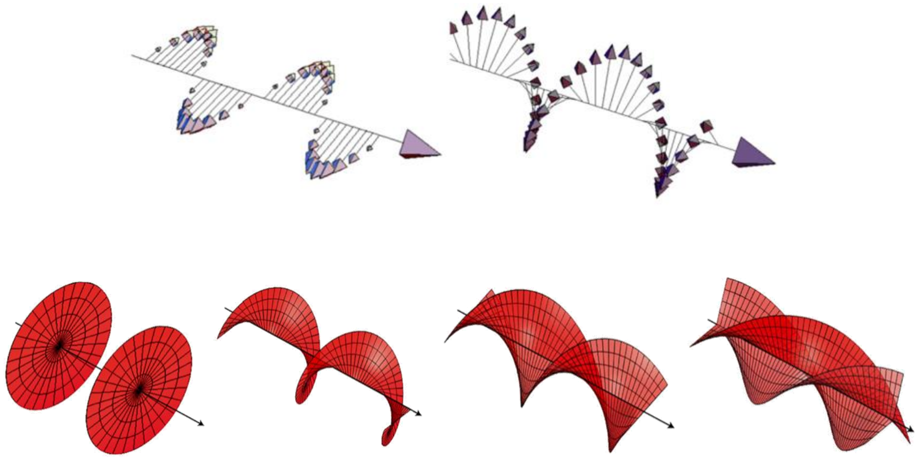

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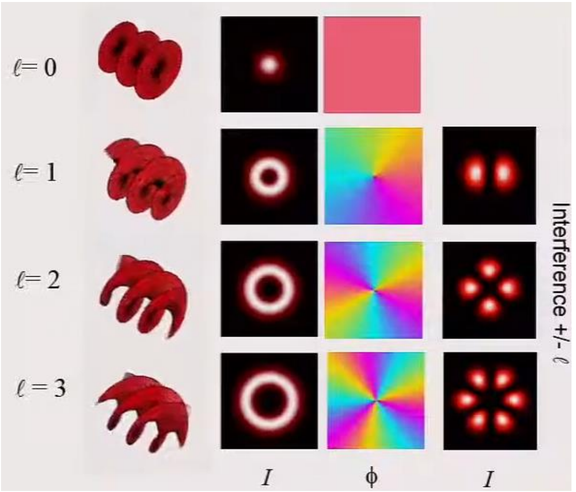

- LG beams and vector vortex beams (transverse plane)

Reproduced from Miles J. Padgett, International School on Parametric Nonlinear Optics (2015)

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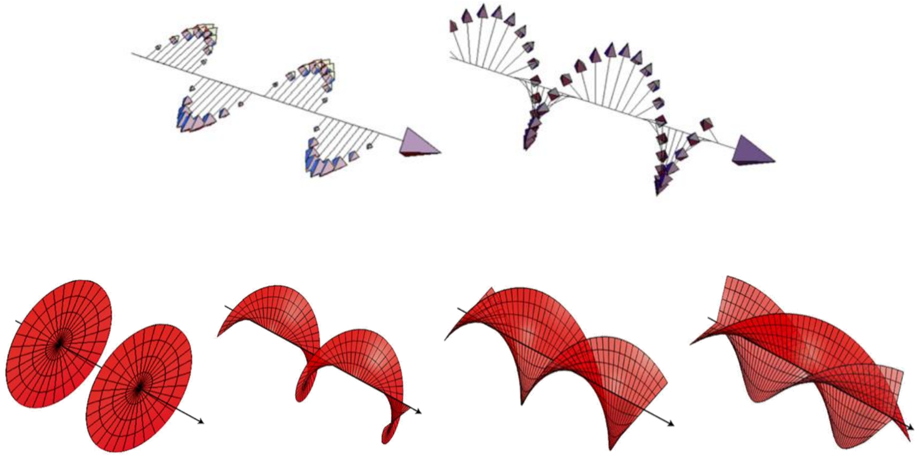

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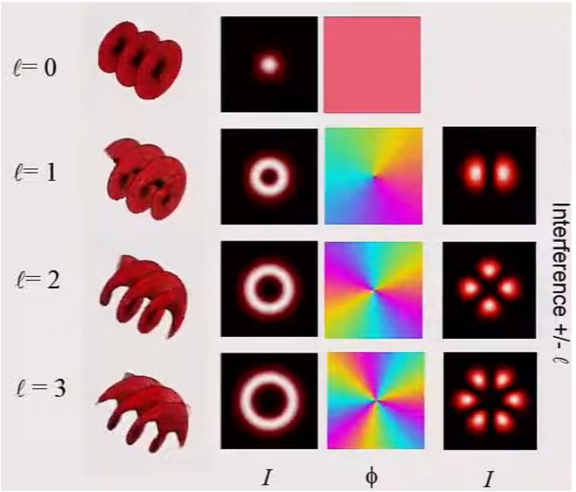

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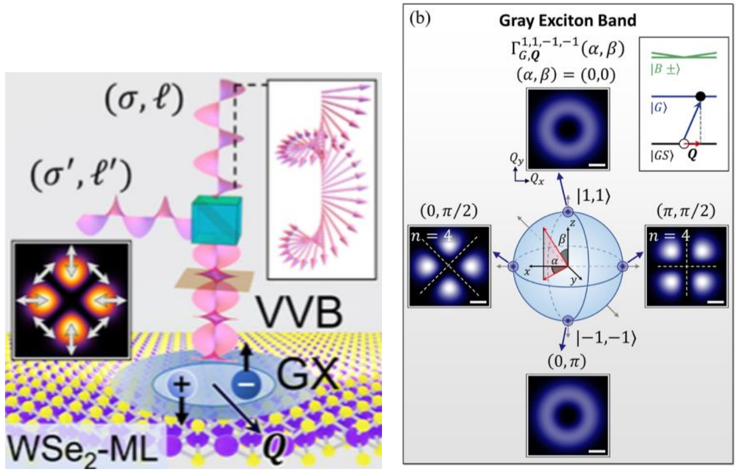
Enhanced Photo-excitation and Angular-Momentum Imprint of Gray Excitons in WSe_2 Monolayers by Spin–Orbit-Coupled Vector Vortex Beams

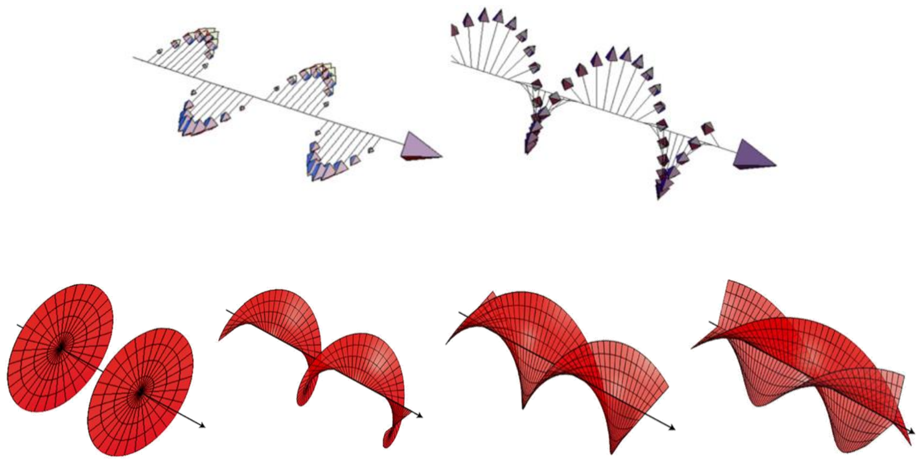
Oscar Javier Gomez Sanchez,[§] Guan-Hao Peng,[*,§] Wei-Hua Li, Ching-Hung Shih, Chao-Hsin Chien, and Shun-Jen Cheng[*]



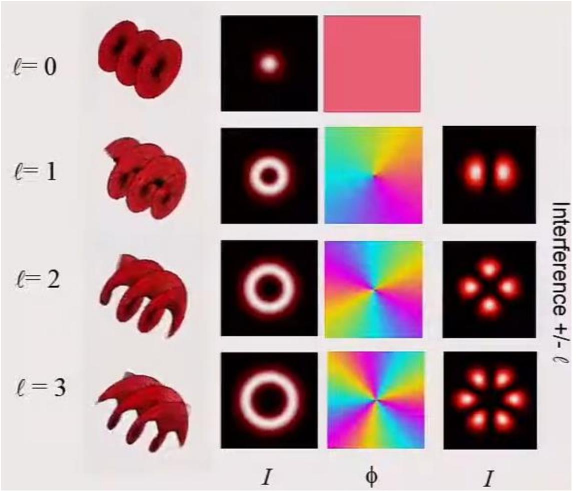
- SOI influence optical transition rate through vector vortex beam
O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024)

1. Introduction

- Angular momentum in light

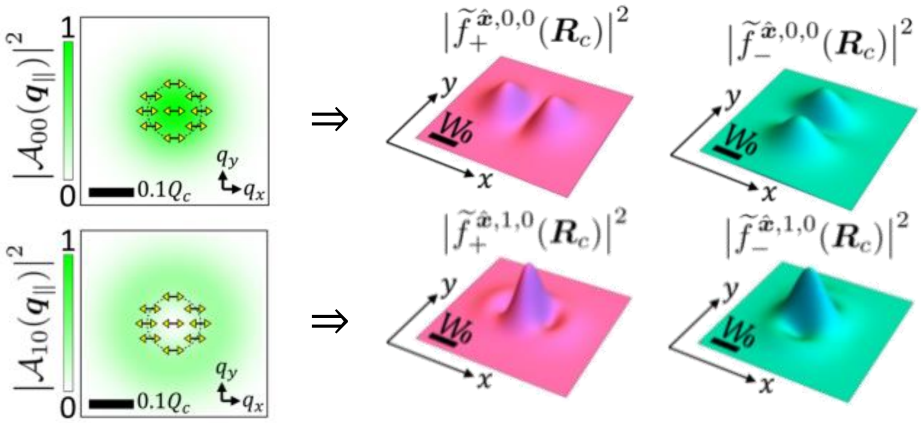


- Polarization of light / Helical phase fronts for $\ell = 0, 1, 2, 3$
A. M. Yao & M. J. Padgett, *Adv. Opt. Photonics* **3**, 161 (2011)

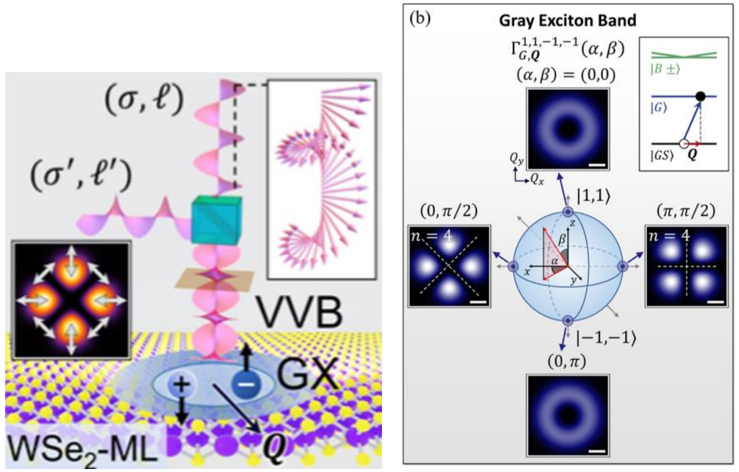


- LG beams and vector vortex beams (transverse plane)
Reproduced from Miles J. Padgett, International School on Parametric Nonlinear Optics (2015)

- Light matter interaction



- OAM can influence exciton's wave packet structures
G.-H. Peng *et al.*, *Nano Lett.* **19**, 2299 (2019)



- SOI influence optical transition rate through vector vortex beam
O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024)

Outline

2. Laguerre-Gaussian (LG) beam : Derivation

LG beam is solved as vector potential $\boldsymbol{A}(\boldsymbol{r})$ from paraxial Helmholtz equation within **Lorentz gauge**.

Maxwell equations

scalar Helmholtz equation

$$(\nabla^2 + q_0^2)A(\boldsymbol{r}) = 0$$

paraxial approximation

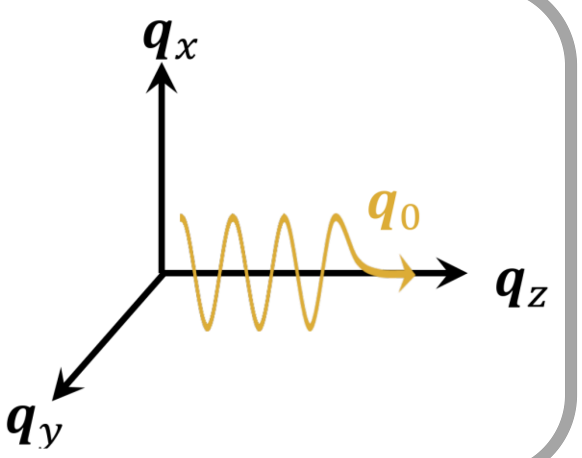
$$\left(\nabla_\parallel^2 + \frac{\partial^2}{\partial z^2} + 2iq_0 \frac{\partial}{\partial z}\right) U(\boldsymbol{r}) = 0$$

Ansatz

Paraxial wave

$$A(\boldsymbol{r}) = U(\boldsymbol{r}) e^{iq_0 z}$$

wave vector $\boldsymbol{q}_z \approx \boldsymbol{q}_0$



Approx.

$$\left|\frac{\partial^2}{\partial z^2} U(\boldsymbol{r})\right| \ll q_0 \left|\frac{\partial}{\partial z} U(\boldsymbol{r})\right|$$

$$\left|\frac{\partial^2}{\partial z^2} U(\boldsymbol{r})\right| \ll \left|\nabla_\parallel^2 U(\boldsymbol{r})\right|$$

Fail

paraxial Helmholtz equation

$$\left(\nabla_\parallel^2 + 2iq_0 \frac{\partial}{\partial z}\right) U(\boldsymbol{r}) \approx 0$$

Coulomb gauge condition

$$\boldsymbol{\nabla} \cdot \boldsymbol{A}^C(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma \cdot \boldsymbol{\nabla}_\parallel A^C(\boldsymbol{r})$$

$$= \left(\frac{\partial}{\partial x} + i\sigma \frac{\partial}{\partial y}\right) A^C(\boldsymbol{r}) = 0, \quad \text{for } \hat{\boldsymbol{\varepsilon}}_\sigma = \hat{\boldsymbol{x}} + i\sigma\hat{\boldsymbol{y}}$$

$$\sigma = \pm 1$$

eventually constraint the transverse component

$$\nabla_\parallel^2 A^C(\boldsymbol{r}) = \left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right) A^C(\boldsymbol{r}) = 0.$$

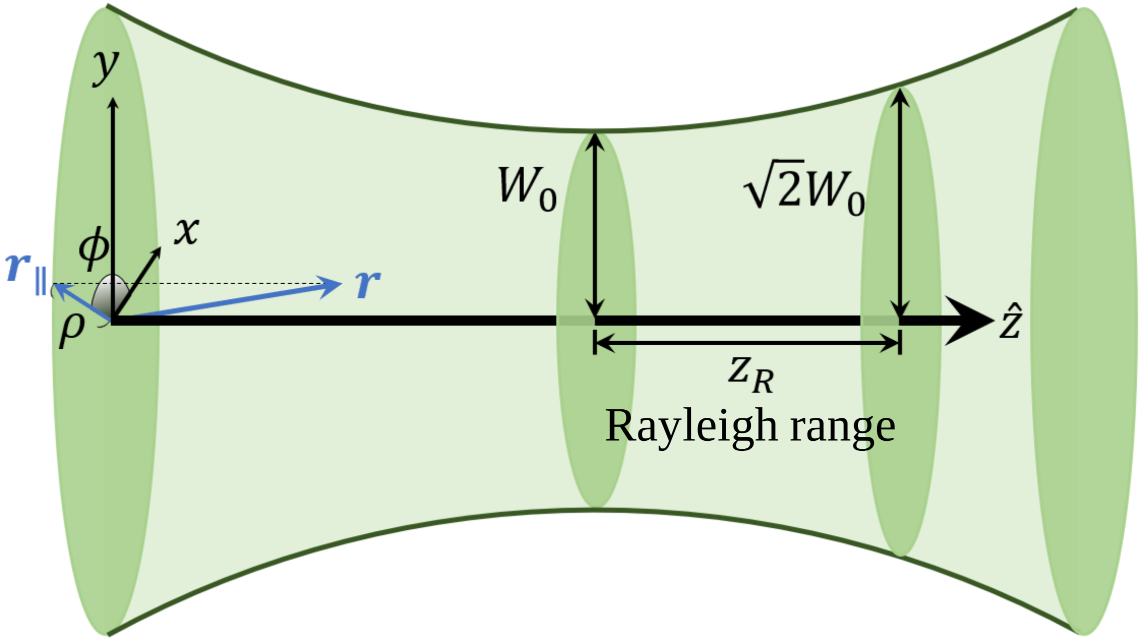
2. Laguerre-Gaussian (LG) beam : Distribution

LG beam is solved as vector potential $\boldsymbol{A}(\boldsymbol{r})$ from paraxial Helmholtz equation within Lorentz gauge.

With **long Rayleigh range approximation** :

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma\, A^{L}_{\ell,p}(\boldsymbol{r}) \approx \hat{\boldsymbol{\varepsilon}}_\sigma\, f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\mathrm{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}}\, L_p^{|\ell|}\left(\frac{2\rho^2}{W_0^2}\right)\left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$



- Polarization unit vector :

$$\hat{\boldsymbol{\varepsilon}}_\sigma = \hat{\boldsymbol{x}} + i\sigma\hat{\boldsymbol{y}}, \quad \sigma = \pm 1$$

(LG beam carries SAM of $\sigma\hbar$)

- Topological charge:

$$\ell \in Z$$

(LG beam carries OAM of $\ell\hbar$)

- Beam waist : W_0
- Rayleigh range : $z_R = \frac{1}{2} q_0 W_0^2$
- Radial mode index: integer $p \geq 0$
- Constant amplitude : A_0^{LG}
- Generalized Laguerre polynomial :

$$L_p^{|\ell|}(x) = \sum_{m=0}^{p} (-1)^m \frac{(|\ell|+p)!}{(p-m)!(|\ell|+m)!m!} x^m$$

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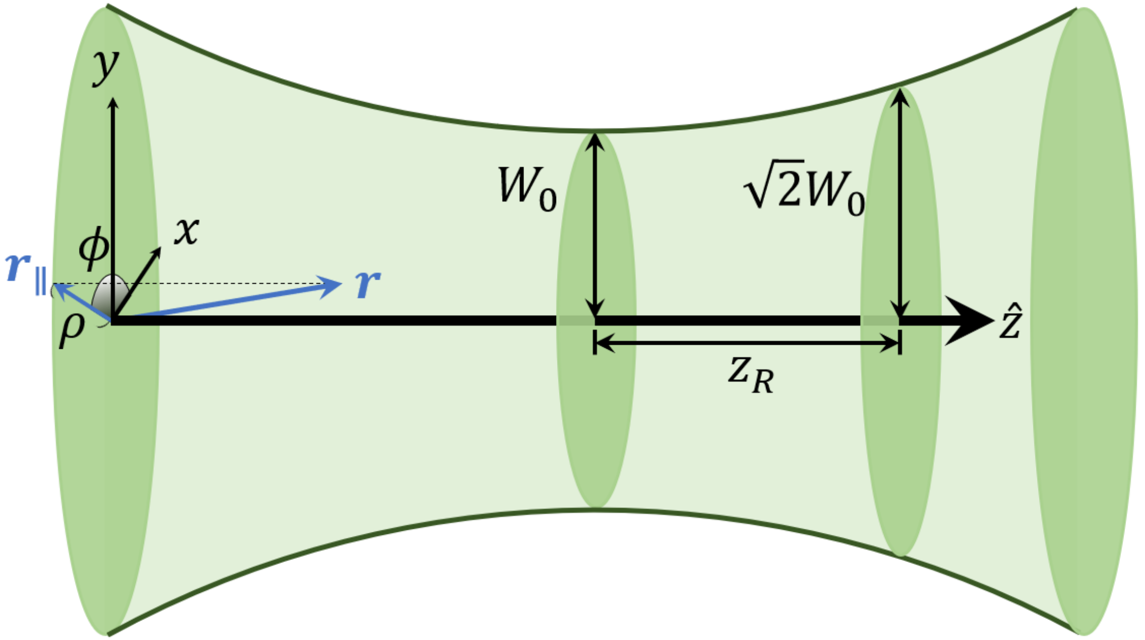

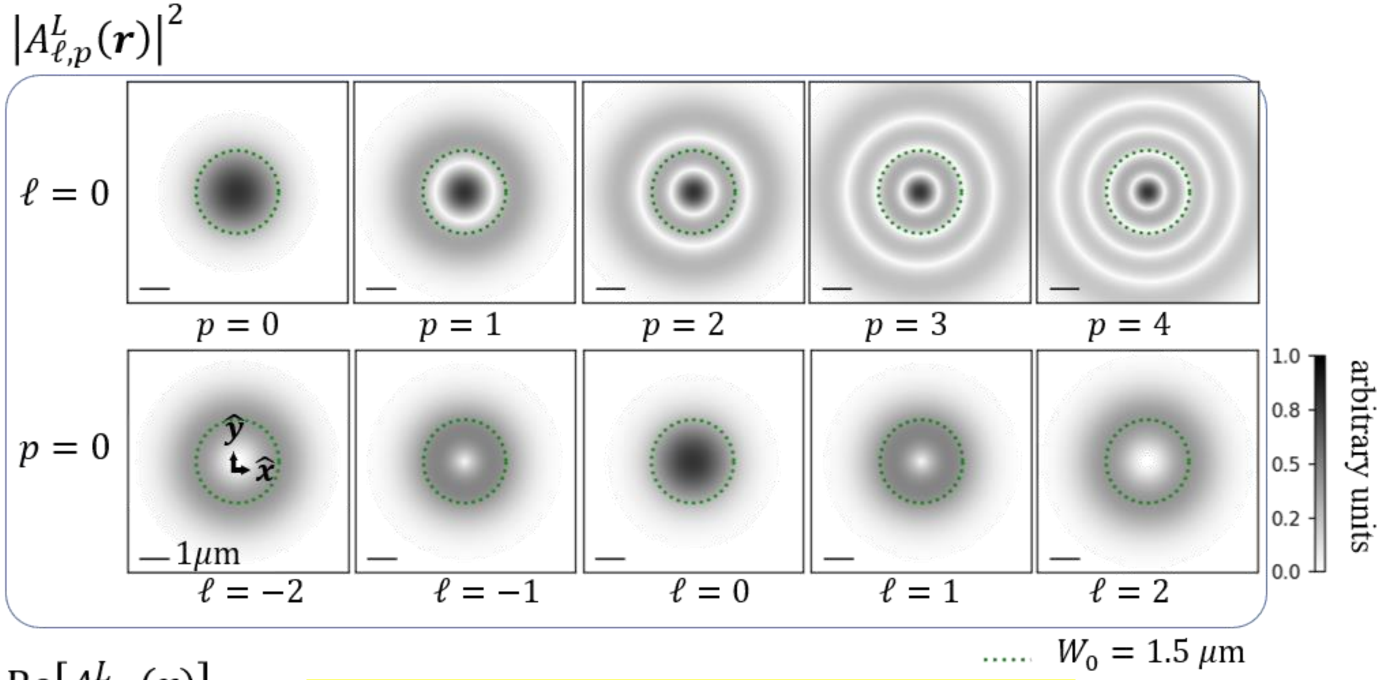

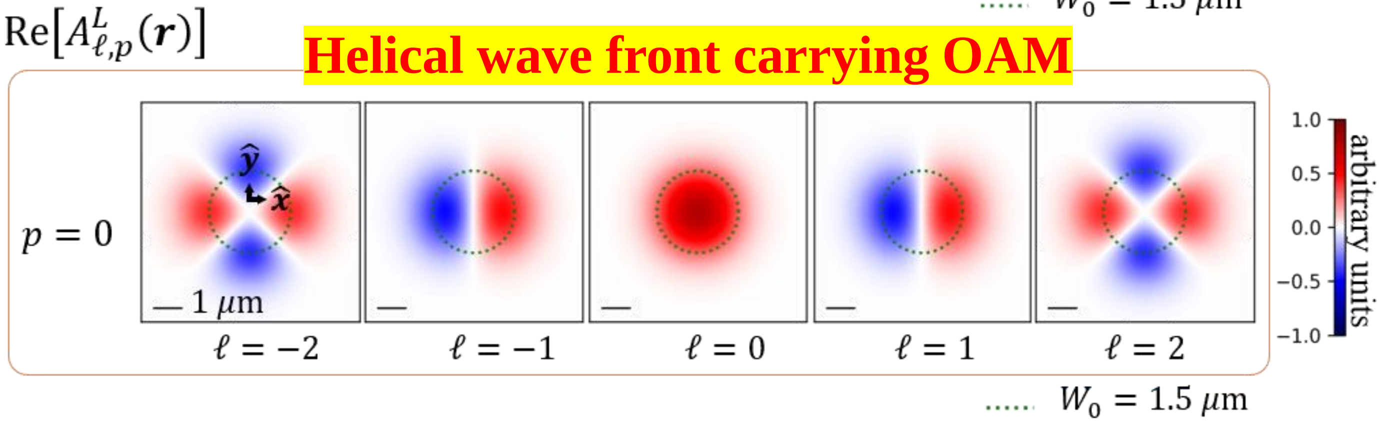

2. Laguerre-Gaussian (LG) beam : Requirement

LG beam in real space / Lorentz gauge : $\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r})$

$\Downarrow$

LG beam in angular spectrum / Coulomb gauge : $\boldsymbol{\mathcal{A}}^{C}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel})$ **What we really need**

$\Downarrow$

Theory of light matter interaction

$$\Gamma^{S,\boldsymbol{Q}} = \frac{2\pi}{\hbar}\left|\widetilde{M}^{S,\boldsymbol{Q}}\right|^{2}\rho^{L}\left(\hbar\omega = E^{S,\boldsymbol{Q}}\right)$$

$$\widetilde{M}^{S,\boldsymbol{Q}} \approx \frac{E_g}{2i\hbar}\delta_{q_{\parallel}\boldsymbol{Q}}\boxed{\boldsymbol{\mathcal{A}}^{C}(\boldsymbol{q}_{\parallel})}\cdot \boldsymbol{D}^{S,\boldsymbol{Q}*}$$

2. Laguerre-Gaussian (LG) beam : Angular spectrum

LG beam in Lorentz gauge / **real space** :

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_{\sigma}\, A^{L}_{\ell,p}(\boldsymbol{r}) \approx \hat{\boldsymbol{\varepsilon}}_{\sigma}\, f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\mathrm{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}}\, L_p^{|\ell|}\left(\frac{2\rho^2}{W_0^2}\right)\left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

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LG beam in Lorentz gauge / **angular spectrum** :

$$\boldsymbol{\mathcal{A}}^{L}_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) = \hat{\boldsymbol{\varepsilon}}_\sigma\, \mathcal{A}^{L}_{\ell,p}(\boldsymbol{q}_\parallel) \approx \hat{\boldsymbol{\varepsilon}}_\sigma \tilde{F}_{|\ell|,p}(q_\parallel) e^{i\ell\phi_{q_\parallel}}$$

$$\tilde{F}_{|\ell|,p}(q_\parallel) = \frac{2\pi(-i)^{|\ell|}}{\Omega}\, \boldsymbol{\mathcal{H}}_{|\ell|}\left[f_{\ell,p}(\rho)\right]$$

Hankel transform of order $|\ell|$

Method

Expanded **in plane-wave basis**

$$\boldsymbol{A}(\boldsymbol{r}) = \sum_{\boldsymbol{q}} \boldsymbol{\mathcal{A}}(\boldsymbol{q})\, e^{i\,\boldsymbol{q}\cdot\boldsymbol{r}}$$

Define Fourier transformation

$$f_{\ell,p}(\rho)e^{i\ell\phi} = u_{\ell,p}\,(\boldsymbol{r}_\parallel) = \frac{1}{(2\pi)^2}\int d^2\boldsymbol{q}_\parallel\, F^{LG}_{\ell,p}(q_\parallel) e^{i\,\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}$$

$$F^{LG}_{\ell,p}(q_\parallel) = \int d^2\boldsymbol{r}_\parallel\, u_{\ell,p}(\boldsymbol{r}_\parallel) e^{-i\,\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}$$

Results

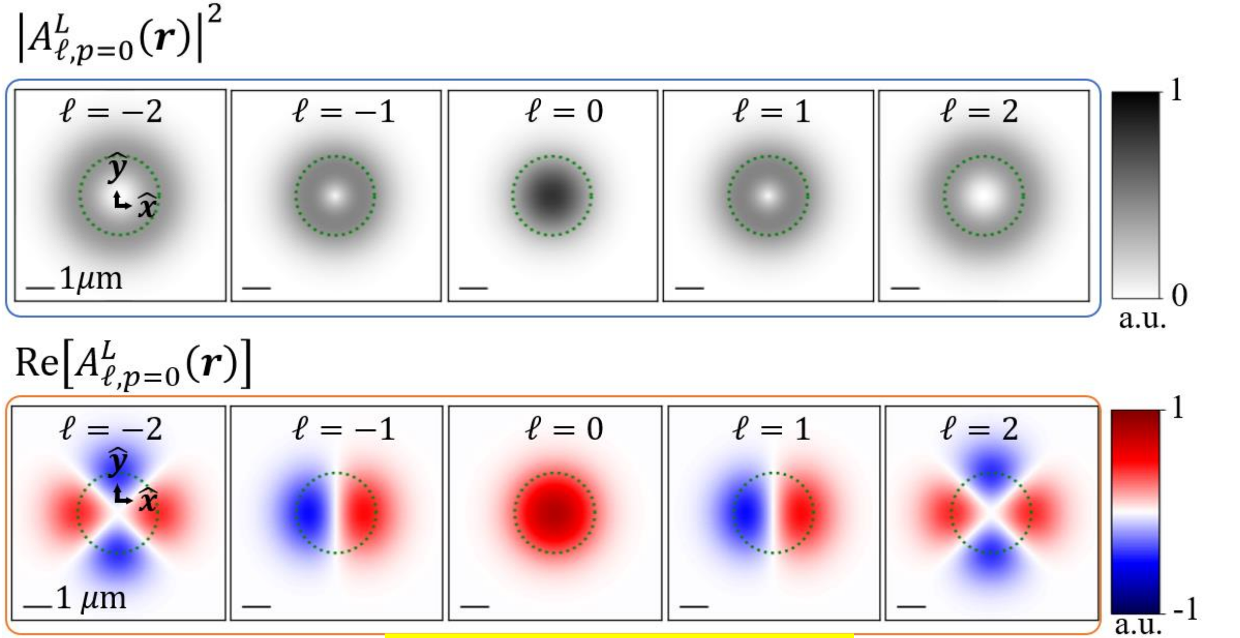
$$\boldsymbol{\mathcal{A}}^{L}_{\sigma,\ell,p}(\mathbf{r}) = \sum_{\boldsymbol{q}_\parallel} \boldsymbol{\mathcal{A}}^{L}_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) e^{i\boldsymbol{q}_\parallel\cdot\mathbf{r}_\parallel} e^{iq_0 z}$$

2. Laguerre-Gaussian (LG) beam : Angular spectrum

LG beam in Lorentz gauge / **real space** :

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma\, A^{L}_{\ell,p}(\boldsymbol{r}) \approx {\color{red}\hat{\boldsymbol{\varepsilon}}_\sigma}\, f_{\ell,p}(\rho){\color{red}e^{i\ell\phi}} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\mathrm{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}}\, L_p^{|\ell|}\left(\frac{2\rho^2}{W_0^2}\right)\left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$



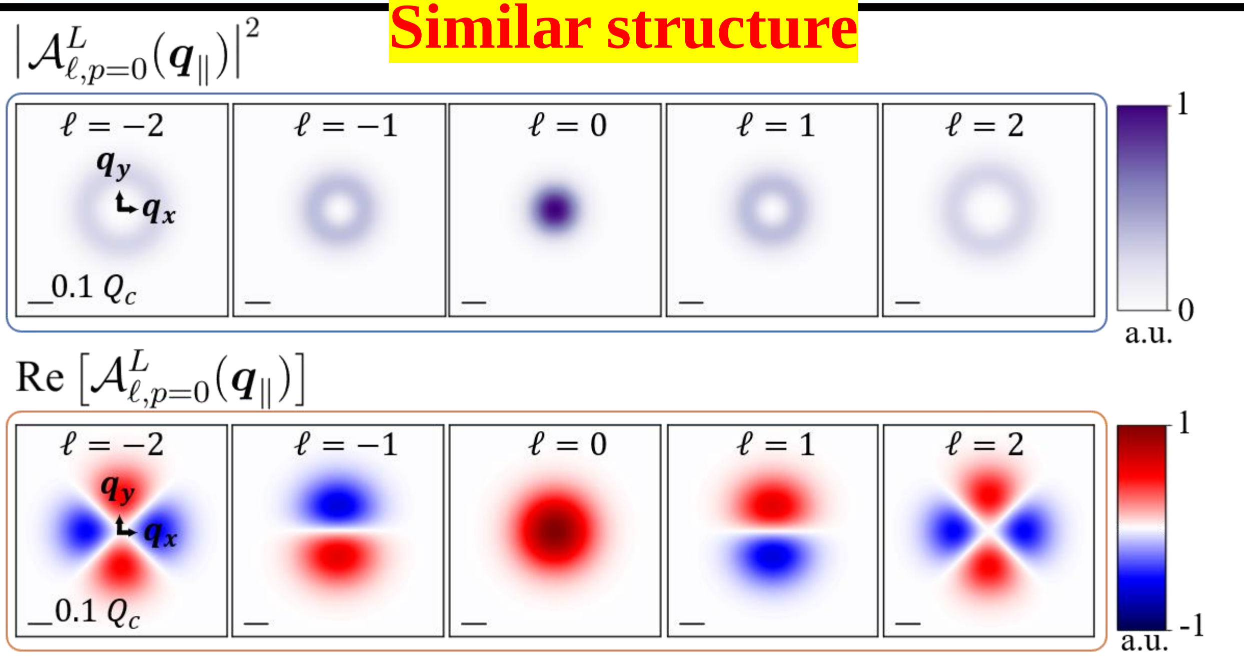
Similar structure

LG beam in Lorentz gauge / **angular spectrum** :

$$\boldsymbol{\mathcal{A}}^{L}_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) = \hat{\boldsymbol{\varepsilon}}_\sigma\, \mathcal{A}^{L}_{\ell,p}(\boldsymbol{q}_\parallel) \approx {\color{red}\hat{\boldsymbol{\varepsilon}}_\sigma} \tilde{F}_{|\ell|,p}(q_\parallel){\color{red}e^{i\ell\phi_{q_\parallel}}}$$

$$\tilde{F}_{|\ell|,p}(q_\parallel) = \frac{2\pi(-i)^{|\ell|}}{\Omega}\, \mathcal{H}_{|\ell|}\left[f_{\ell,p}(\rho)\right]$$

Hankel transform of order $|\ell|$



2. Laguerre-Gaussian (LG) beam : Coulomb gauge

LG beam in **Lorentz gauge** / real space :

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_{\sigma}\, A^{L}_{\ell,p}(\boldsymbol{r}) \approx \hat{\boldsymbol{\varepsilon}}_{\sigma}\, f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

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LG beam in **Coulomb gauge /** real space:

$$\boldsymbol{A}_{\sigma,\ell,p}(r) \equiv \boldsymbol{A}^{C}_{\sigma,\ell,p}(r) \approx \tilde{A}^{\parallel}_{\ell,p}(r)\hat{\boldsymbol{\varepsilon}}_\sigma + \tilde{A}^{z}_{\sigma,\ell,p}(r)\hat{\boldsymbol{z}}$$

Transverse component :

$$\tilde{A}^{\parallel}_{\ell,p}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Longitudinal component :

$$\tilde{A}^{z}_{\sigma,\ell,p}(r) \approx \frac{i}{\sqrt{2}q_0\rho}\left[(|\ell|+\sigma\ell) - \frac{2\rho^2}{W_0^2}\right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

Additional component

Method

Relation from Maxwell's equations

$$\boldsymbol{A}^{C}_{\sigma,\ell,p}(r) = \boldsymbol{A}^{L}_{\sigma,\ell,p}(r) + \frac{\nabla\left(\nabla\cdot \boldsymbol{A}^{L}_{\sigma,\ell,p}(r)\right)}{q_0^2}$$

2. Laguerre-Gaussian (LG) beam : Coulomb gauge

LG beam in **Lorentz gauge** / real space :

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma\, A^{L}_{\ell,p}(\boldsymbol{r}) \approx \hat{\boldsymbol{\varepsilon}}_\sigma\, f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

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LG beam in **Coulomb gauge / **real space:

$$\boldsymbol{A}_{\sigma,\ell,p}(r) \equiv \boldsymbol{A}^{C}_{\sigma,\ell,p}(r) \approx \tilde{A}^{\parallel}_{\ell,p}(r)\hat{\boldsymbol{\varepsilon}}_\sigma + \tilde{A}^{z}_{\sigma,\ell,p}(r)\hat{\boldsymbol{z}}$$

Transverse component :

$$\tilde{A}^{\parallel}_{\ell,p}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Longitudinal component :

SOI

$$\tilde{A}^{z}_{\sigma,\ell,p}(r) \approx \frac{i}{\sqrt{2}q_0\rho}\left[(|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2}\right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

2. Laguerre-Gaussian (LG) beam : SOI appear

LG beam in **Lorentz gauge** / real space :

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma\, A^{L}_{\ell,p}(\boldsymbol{r}) \approx \hat{\boldsymbol{\varepsilon}}_\sigma\, f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

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LG beam in **Coulomb gauge** / real space:

$$\boldsymbol{A}_{\sigma,\ell,p}(r) \equiv \boldsymbol{A}^{C}_{\sigma,\ell,p}(r) \approx \tilde{A}^{\parallel}_{\ell,p}(r)\hat{\boldsymbol{\varepsilon}}_\sigma + \tilde{A}^{z}_{\sigma,\ell,p}(r)\hat{\boldsymbol{z}}$$

Transverse component :

$$\tilde{A}^{\parallel}_{\ell,p}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Longitudinal component :

SOI

$$\tilde{A}^{z}_{\sigma,\ell,p}(r) \approx \frac{i}{\sqrt{2}q_0\rho}\left[(|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2}\right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

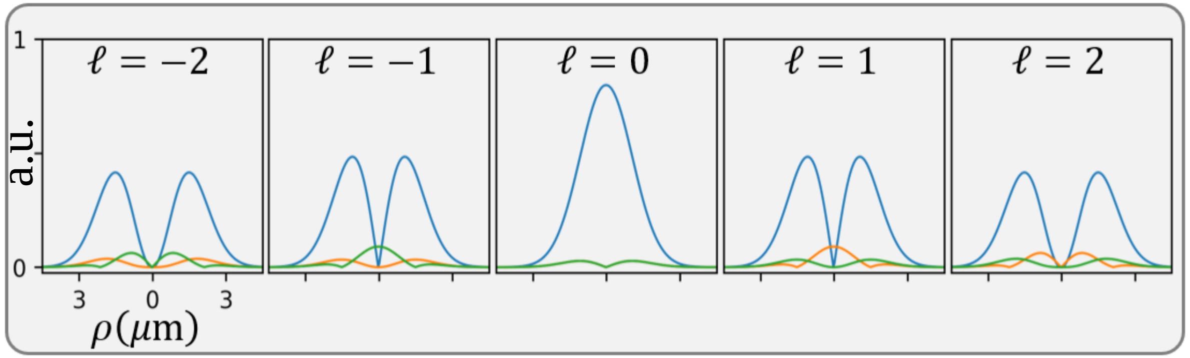
— $|\tilde{A}^{\parallel}_{\ell,p=0}(r)|$

— $|\tilde{A}^{z}_{\sigma=1,\ell,p=0}(r)|$

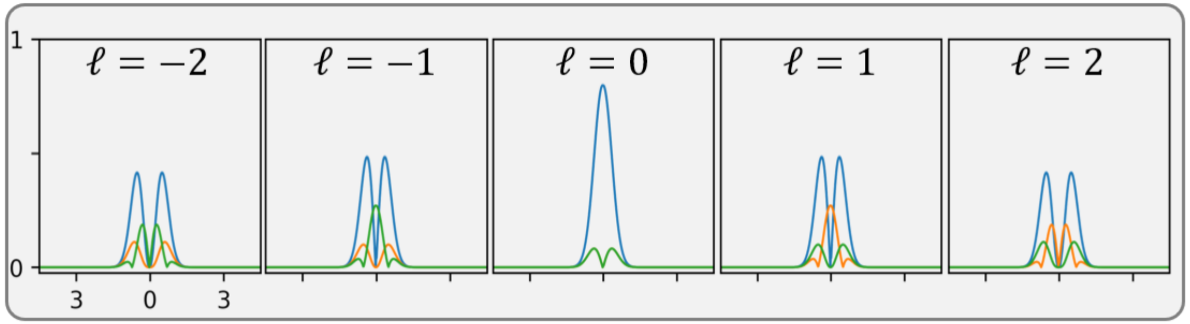
— $|\tilde{A}^{z}_{\sigma=-1,\ell,p=0}(r)|$

	Transverse	**Longitudinal**
$\pm\ell$	symmetric	Asymmetric **SOI**
$W_0 \downarrow$	merely changed	Larger

$W_0 = 1.5\mu\mathrm{m}$



$W_0 = 0.5\mu\mathrm{m}$



2. Longitudinal component : SOI effect

LG beam solutions:

Real / Lorentz :

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_{\sigma}\, A^{L}_{\ell,p}(\boldsymbol{r}) \approx \hat{\boldsymbol{\varepsilon}}_{\sigma}\, f_{\ell,p}(\rho)e^{i\ell\phi}e^{iq_0 z}$$

Angular / Lorentz :

$$\boldsymbol{\mathcal{A}}^{L}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel}) = \hat{\boldsymbol{\varepsilon}}_{\sigma}\, \mathcal{A}^{L}_{\ell,p}(\boldsymbol{q}_{\parallel}) \approx \hat{\boldsymbol{\varepsilon}}_{\sigma}\tilde{F}_{|\ell|,p}(q_{\parallel})e^{i\ell\phi_{\boldsymbol{q}_{\parallel}}}$$

Real / Coulomb :

$$\boldsymbol{A}_{\sigma,\ell,p}(r) \equiv \boldsymbol{A}^{C}_{\sigma,\ell,p}(r) \approx \tilde{A}^{\parallel}_{\ell,p}(r)\hat{\boldsymbol{\varepsilon}}_{\sigma} + \tilde{A}^{z}_{\sigma,\ell,p}(r)\hat{\boldsymbol{z}}$$

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Real / Coulomb :

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Method

Expanded in plane-wave basis

$$\boldsymbol{A}^{C}_{\sigma,\ell,p}(r) = \frac{1}{(2\pi)^2}\int \mathrm{d}\boldsymbol{q}_{\parallel}\, \boldsymbol{\mathcal{A}}^{C}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel}) e^{i\,\boldsymbol{q}\cdot\boldsymbol{r}}$$

$$\boldsymbol{\mathcal{A}}^{C}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel})$$

What we really need

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_{\parallel}) \equiv \boldsymbol{\mathcal{A}}^{C}_{\sigma,\ell,p=0}(\boldsymbol{q}_{\parallel}) \approx \tilde{\mathcal{A}}^{\parallel}_{\ell,p}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{\varepsilon}}_\sigma + \tilde{\mathcal{A}}^{z}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{z}}$$

Transverse component :

$$\tilde{\mathcal{A}}^{\parallel}_{\ell}(\boldsymbol{q}_{\parallel}) \equiv \tilde{\mathcal{A}}^{\parallel}_{\ell,p=0}(\boldsymbol{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{\boldsymbol{q}_{\parallel}}}$$

Longitudinal component :

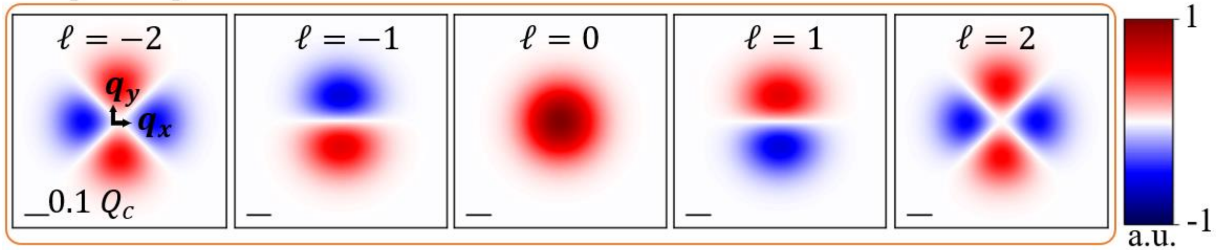
$$\tilde{\mathcal{A}}^{z}_{\sigma,\ell}(\boldsymbol{q}_{\parallel}) \equiv \tilde{\mathcal{A}}^{z}_{\sigma,\ell,p=0}(\boldsymbol{q}_{\parallel}) \approx -\frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{\boldsymbol{q}_{\parallel}}}$$

SOI

2. Longitudinal component : SOI effect

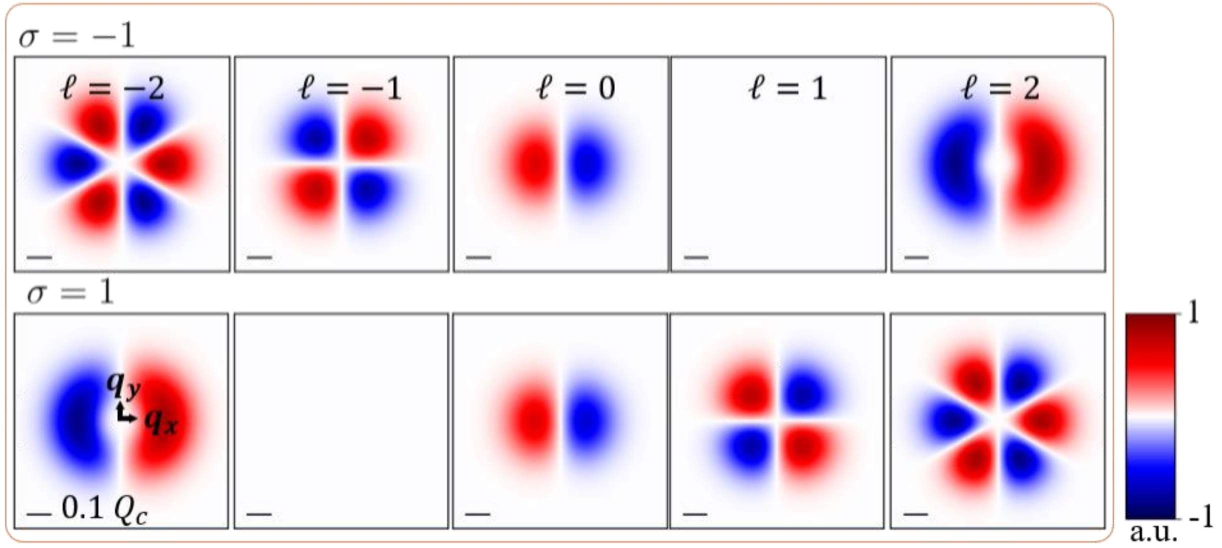
Transverse component :

$\mathrm{Re}\left[\tilde{\mathcal{A}}^{\parallel}_{\ell}(\boldsymbol{q}_{\parallel})\right]$



Longitudinal component :

$\mathrm{Re}\left[\tilde{\mathcal{A}}^{z}_{\sigma,\ell}(\boldsymbol{q}_{\parallel})\right]$



$$\mathcal{A}^{C}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel})$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\mathcal{A}_{\sigma,\ell}(\boldsymbol{q}_{\parallel}) \equiv \mathcal{A}^{C}_{\sigma,\ell,p=0}(\boldsymbol{q}_{\parallel}) \approx \tilde{\mathcal{A}}^{\parallel}_{\ell,p}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{\varepsilon}}_{\sigma} + \tilde{\mathcal{A}}^{z}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{z}}$$

Transverse component :

$$\tilde{\mathcal{A}}^{\parallel}_{\ell}(\boldsymbol{q}_{\parallel}) \equiv \tilde{\mathcal{A}}^{\parallel}_{\ell,p=0}(\boldsymbol{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel})e^{i\ell\phi_{\boldsymbol{q}_{\parallel}}}$$

Longitudinal component :

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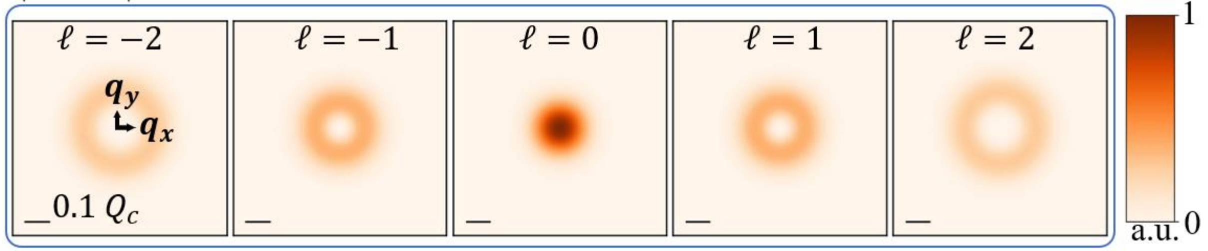
SOI

2. Longitudinal component : SOI effect

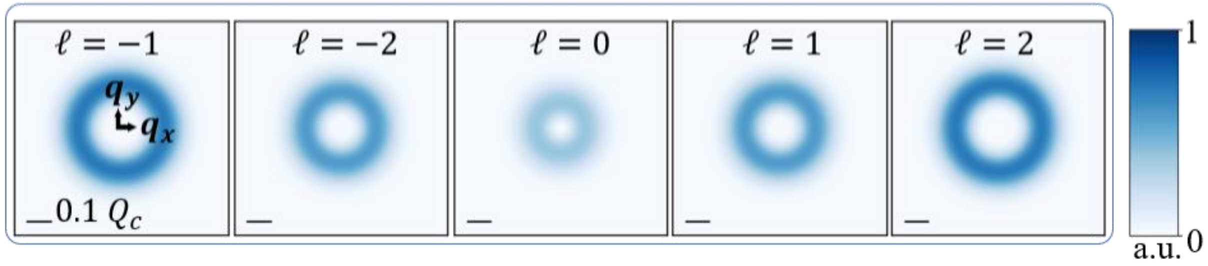
Transverse component :

$\left|\tilde{\mathcal{A}}^{\parallel}_{\ell}(\boldsymbol{q}_{\parallel})\right|^2$



Longitudinal component :

$\left|\tilde{\mathcal{A}}^{z}_{\sigma,\ell}(\boldsymbol{q}_{\parallel})\right|^2$



$$\boldsymbol{\mathcal{A}}^{C}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel})$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_{\parallel}) \equiv \boldsymbol{\mathcal{A}}^{C}_{\sigma,\ell,p=0}(\boldsymbol{q}_{\parallel}) \approx \tilde{\mathcal{A}}^{\parallel}_{\ell,p}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{\varepsilon}}_{\sigma} + \tilde{\mathcal{A}}^{z}_{\sigma,\ell,p}(\boldsymbol{q}_{\parallel})\hat{\boldsymbol{z}}$$

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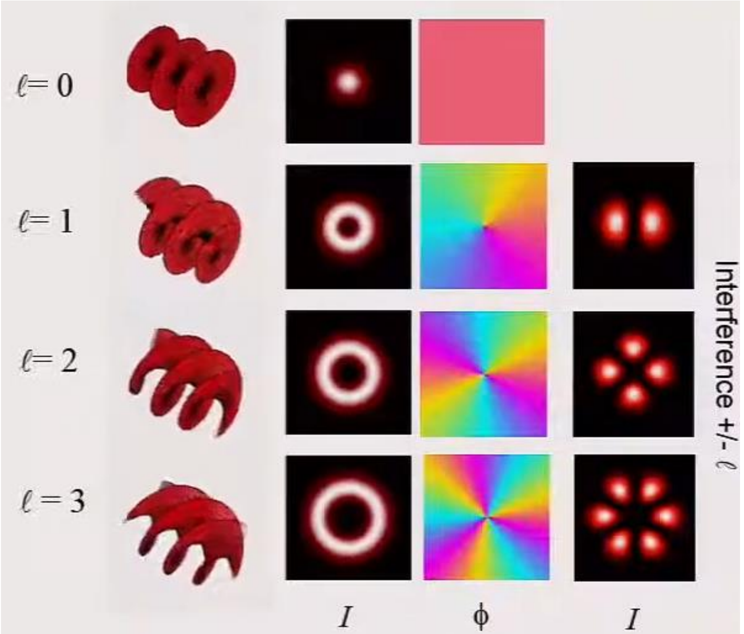
SOI

2. Vector vortex beams (VVB)

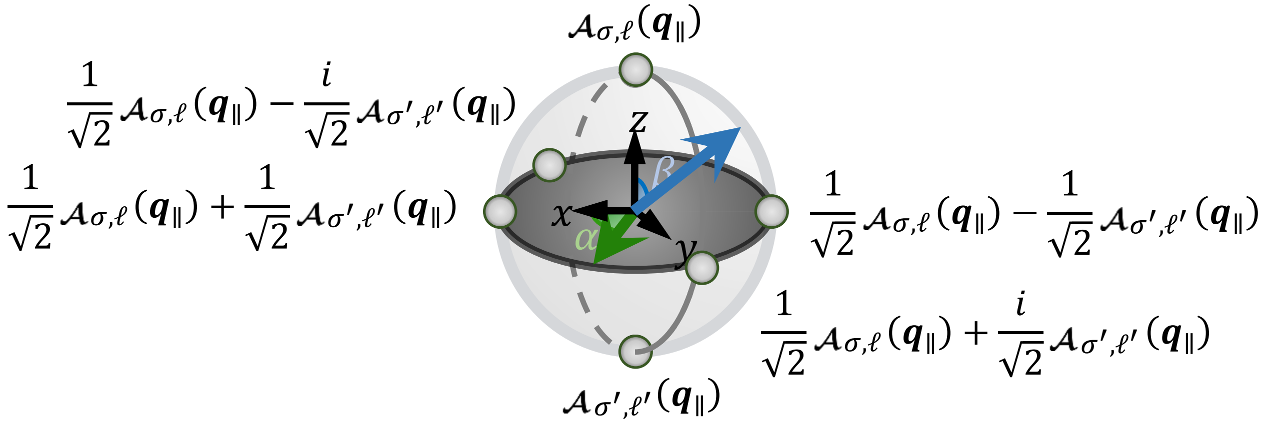
Forming vector vortex beams (VVB) by Superimposing LG beams

$$\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) = \cos\left(\frac{\beta}{2}\right)\mathcal{A}_{\sigma,\ell}(\boldsymbol{q}_\parallel) + e^{i\alpha}\sin\left(\frac{\beta}{2}\right)\mathcal{A}_{\sigma',\ell'}(\boldsymbol{q}_\parallel)$$



- LG beams and vector vortex beams (transverse plane)

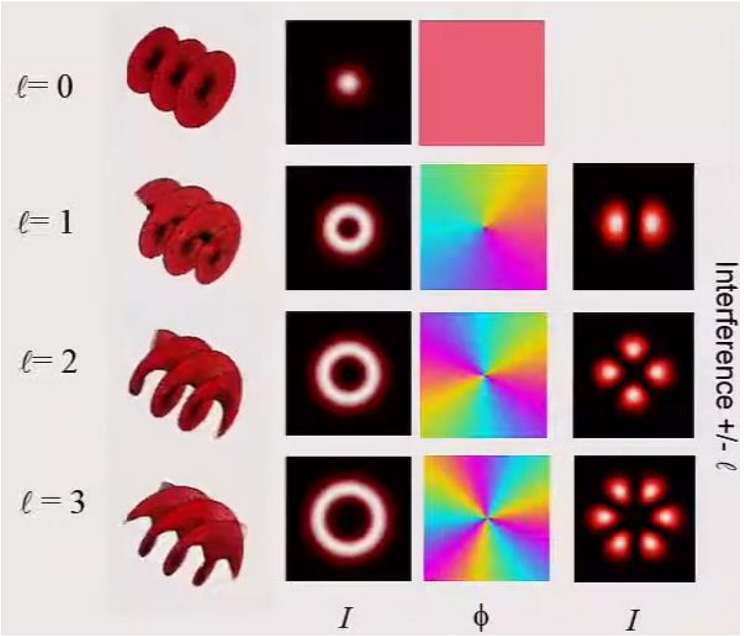
Reproduced from Miles J. Padgett, International School on Parametric Nonlinear Optics (2015)



- Poincaré sphere for VVB states

2. Vector vortex beams (VVB)

Forming vector vortex beams (VVB) by Superimposing LG beams



- LG beams and vector vortex beams (transverse plane)

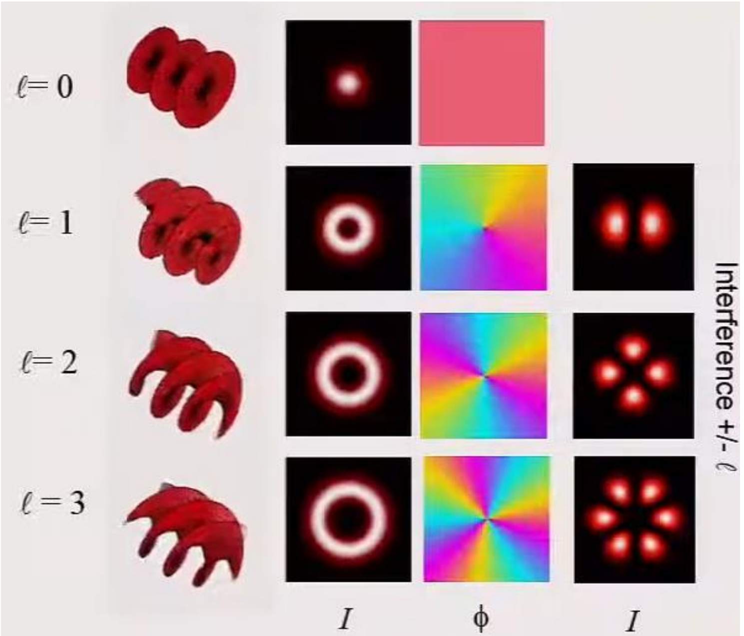
Reproduced from Miles J. Padgett, International School on Parametric Nonlinear Optics (2015)

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta) = \cos\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_{\parallel}) + e^{i\alpha}\sin\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma',\ell'}(\boldsymbol{q}_{\parallel})$$

$$= \boldsymbol{\mathcal{A}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta) + \boldsymbol{\mathcal{A}}^{\boldsymbol{z}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta)$$

2. Vector vortex beams (VVB)

Forming vector vortex beams (VVB) by Superimposing LG beams



- LG beams and vector vortex beams (transverse plane)

Reproduced from Miles J. Padgett, International School on Parametric Nonlinear Optics (2015)

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta) = \cos\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_{\parallel}) + e^{i\alpha}\sin\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma',\ell'}(\boldsymbol{q}_{\parallel})$$

$$= \boldsymbol{\mathcal{A}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta) + \boldsymbol{\mathcal{A}}^{\boldsymbol{z}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta)$$

Define $\Delta\ell \equiv \ell' - \ell \quad 2\bar{\ell} \equiv \ell' + \ell \quad \Delta\sigma \equiv \sigma' - \sigma \quad 2\bar{\sigma} \equiv \sigma' + \sigma$

$$= \boldsymbol{\mathcal{A}}^{\parallel}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta) + \boldsymbol{\mathcal{A}}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta)$$

2. Transverse component : Vector vortex beams (VVB)

Transverse component of VVB:

$$\boxed{\boldsymbol{\mathcal{A}}^{\parallel}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta) = \boldsymbol{\mathcal{A}}^{\boldsymbol{x}}_{\Delta\ell,\bar{\ell}}(\boldsymbol{q}_{\parallel};\alpha,\beta)\widehat{\boldsymbol{x}} + \boldsymbol{\mathcal{A}}^{\boldsymbol{y}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta)\widehat{\boldsymbol{y}}}$$

2. Transverse component : Vector vortex beams (VVB)

Transverse component of VVB:

$$\boldsymbol{\mathcal{A}}^{\parallel}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta) = \boldsymbol{\mathcal{A}}^{\boldsymbol{x}}_{\Delta\ell,\bar{\ell}}(\boldsymbol{q}_{\parallel};\alpha,\beta)\hat{\boldsymbol{x}} + \boldsymbol{\mathcal{A}}^{\boldsymbol{y}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta)\hat{\boldsymbol{y}}$$

Transverse component of VVB: $\Delta\sigma = -2, \bar{\sigma} = 0 \,/\, \beta = \frac{\pi}{2}$

$$\begin{aligned}\boldsymbol{\mathcal{A}}^{\parallel}_{\Delta\ell,\bar{\ell},-2,0}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) &= \frac{1}{2}e^{i\left(\bar{\ell}\phi_{q_\parallel}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}\left(\Delta\ell\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_\parallel)+\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_\parallel)\right]-i\sin\left[\frac{1}{2}\left(\Delta\ell\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_\parallel)-\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_\parallel)\right]\right\}\hat{\boldsymbol{x}} \\ &+\frac{1}{2}e^{i\left(\bar{\ell}\phi_{q_\parallel}+\frac{\alpha}{2}\right)}\left\{\sin\left[\frac{1}{2}\left(\Delta\ell\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_\parallel)+\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_\parallel)\right]+i\sin\left[\frac{1}{2}\left(\Delta\ell\phi_{q_\parallel}+\alpha+\pi\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_\parallel)-\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_\parallel)\right]\right\}\hat{\boldsymbol{y}}\end{aligned}$$

2. Transverse component : Vector vortex beams (VVB)

Transverse component of VVB:

$$\boldsymbol{\mathcal{A}}^{\parallel}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta)=\boldsymbol{\mathcal{A}}^{x}_{\Delta\ell,\bar{\ell}}(\boldsymbol{q}_{\parallel};\alpha,\beta)\hat{\boldsymbol{x}}+\boldsymbol{\mathcal{A}}^{y}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta)\hat{\boldsymbol{y}}$$

Transverse component of VVB: $\Delta\sigma=-2,\bar{\sigma}=0\ /\ \beta=\frac{\pi}{2}$

$$\boldsymbol{\mathcal{A}}^{\parallel}_{\Delta\ell,\bar{\ell},-2,0}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right)=\frac{1}{2}e^{i\left(\bar{\ell}\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}}+\alpha)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel})+\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]-i\sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}}+\alpha)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel})-\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]\right\}\hat{\boldsymbol{x}}$$
$$+\frac{1}{2}e^{i\left(\bar{\ell}\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)}\left\{\sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}}+\alpha)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel})+\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]+i\sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}}+\alpha+\pi)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel})-\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]\right\}\hat{\boldsymbol{y}}$$

If $\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel})=\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})$

$$\boldsymbol{\mathcal{A}}^{\parallel}_{\Delta\ell,\bar{\ell},-2,0}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right)=\boxed{\tilde{F}_{\left|\bar{\ell}\pm\frac{\Delta\ell}{2}\right|}(q_{\parallel})}\,e^{i\left(\bar{\ell}\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)}\boxed{\left\{\cos\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}}+\alpha)\right]\hat{\boldsymbol{x}}+\sin\left[\frac{1}{2}(\Delta\ell\phi_{q_{\parallel}}+\alpha)\right]\hat{\boldsymbol{y}}\right\}}$$

Doughnut shape **Linear polarization**

2. Transverse component : Vector vortex beams (VVB)

Transverse component of VVB:

$$\mathcal{A}^{\parallel}_{\Delta\ell,\bar{\ell},-2,0}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right)=\boxed{\tilde{F}_{\left|\bar{\ell}\pm\frac{\Delta\ell}{2}\right|}(q_{\parallel})}\,e^{i\left(\bar{\ell}\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)}\boxed{\left\{\cos\left[\frac{1}{2}\left(\Delta\ell\phi_{q_{\parallel}}+\alpha\right)\right]\hat{\boldsymbol{x}}+\sin\left[\frac{1}{2}\left(\Delta\ell\phi_{q_{\parallel}}+\alpha\right)\right]\hat{\boldsymbol{y}}\right\}}$$

Doughnut shape **Linear polarization**

Conditions for $\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel})=\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})$ in LG beams

$$\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|=|\ell|=|\ell'|=\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|$$

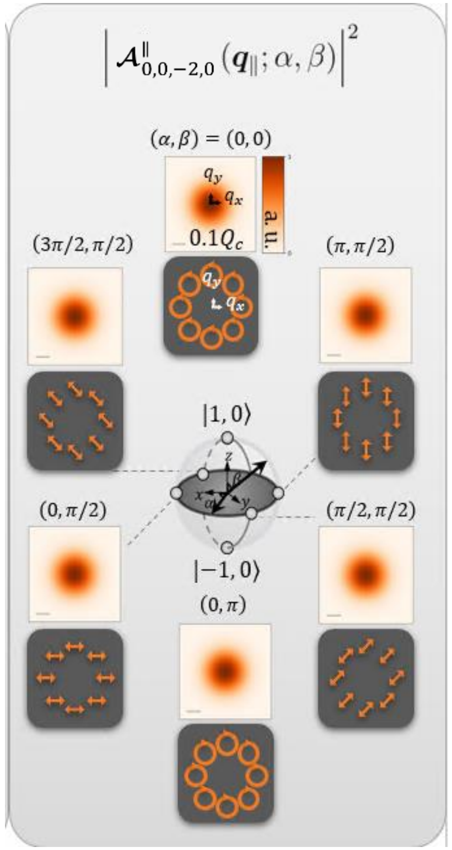
1. $\ell=\ell'$
2. $\ell=-\ell'$

2. Transverse component : Vector vortex beams (VVB)

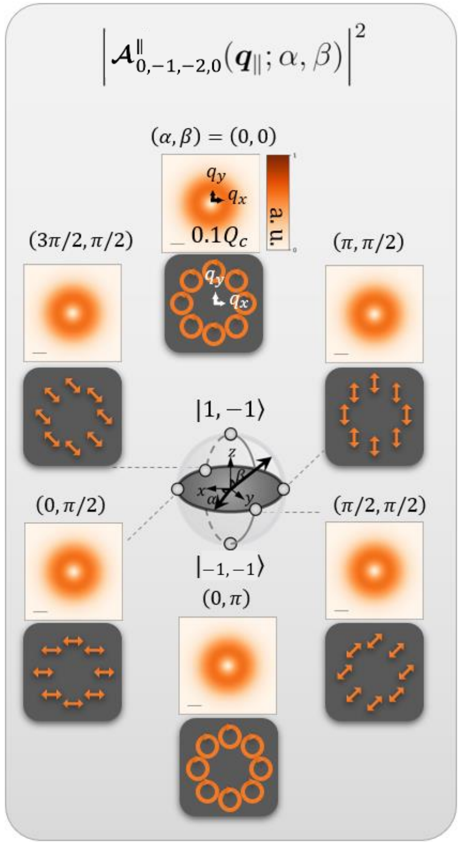
First case : $\ell = \ell'$

$$\mathcal{A}^{\parallel}_{0,\ell,-2,0}\left(\boldsymbol{q}_\parallel;\alpha,\beta=\frac{\pi}{2}\right)=\boxed{\tilde{F}_{|\ell|}(q_\parallel)}\,e^{i\left(\ell\phi_{q_\parallel}+\frac{\alpha}{2}\right)}\boxed{\left\{\cos\left(\frac{\alpha}{2}\right)\hat{\boldsymbol{x}}+\sin\left(\frac{\alpha}{2}\right)\hat{\boldsymbol{y}}\right\}}$$

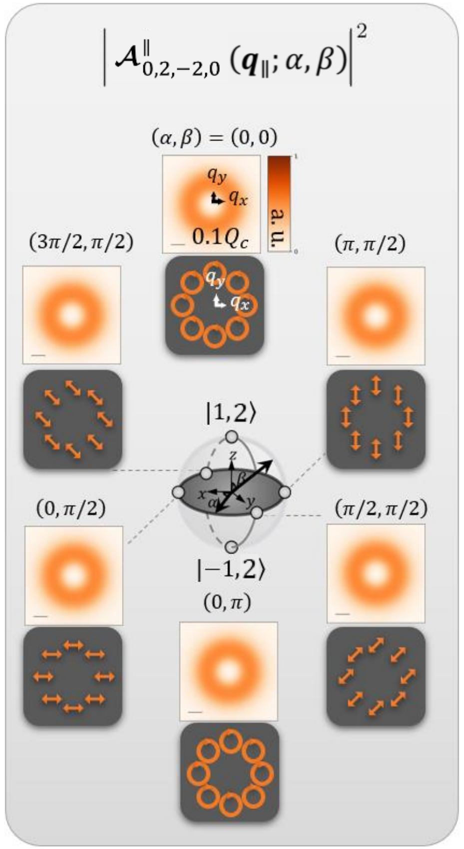
Doughnut shape **Linear polarization changed with $\boldsymbol{\alpha}$**



$\ell = 0$



$\ell = -1$



$\ell = 2$

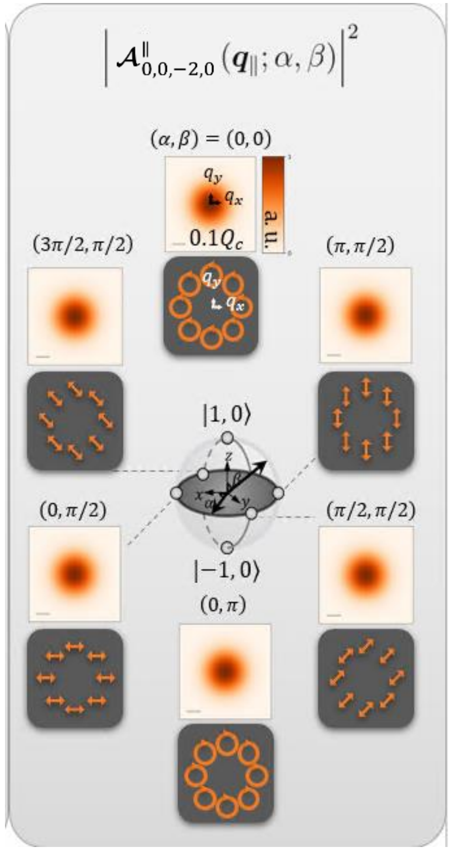
$\sigma,\ell,\sigma',\ell' = 1,2,-1,2$

2. Transverse component : Vector vortex beams (VVB)

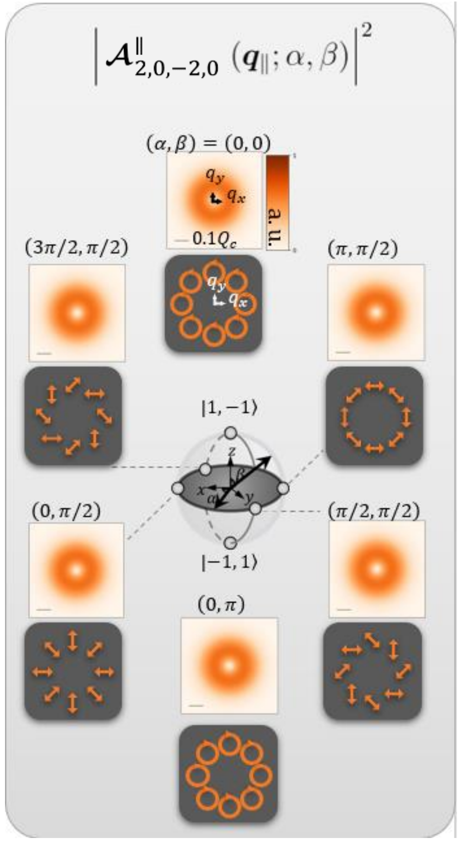
Second case : $\ell = -\ell'$

$$\mathcal{A}^{\parallel}_{-2\ell,0,-2,0}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right)=\boxed{\tilde{F}_{|\ell|}(q_{\parallel})}\,e^{i\frac{\alpha}{2}}\boxed{\left\{\cos\left(-\ell\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)\widehat{\boldsymbol{x}}+\sin\left(-\ell\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)\widehat{\boldsymbol{y}}\right\}}$$

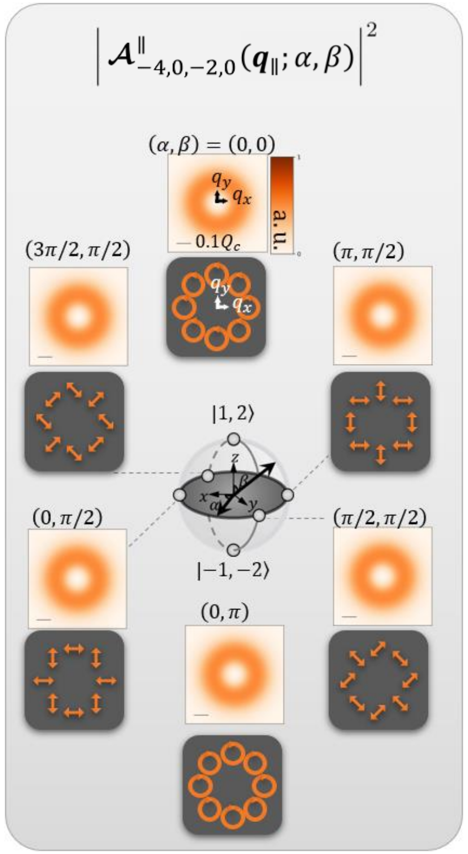
Doughnut shape **Linear polarization changed with $\boldsymbol{\alpha}$ and ℓ**



$\ell = 0$



$\ell = -1$



$\ell = 2$

2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:

$$\boldsymbol{\mathcal{A}}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_\parallel;\alpha,\beta) = -\frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\left((\bar{\ell}+\bar{\sigma})\phi_{q_\parallel}+\frac{\alpha}{2}\right)} \Bigg\{\cos\left[\frac{1}{2}\Big((\Delta\ell+\Delta\sigma)\phi_{q_\parallel}+\alpha\Big)\right]\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_\parallel)+\sin\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_\parallel)\right]$$
$$-\,i\sin\left[\frac{1}{2}\Big((\Delta\ell+\Delta\sigma)\phi_{q_\parallel}+\alpha\Big)\right]\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_\parallel)-\sin\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_\parallel)\right]\Bigg\}\hat{\boldsymbol{z}}$$

2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:

$$\mathcal{A}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta) = -\frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\left((\bar{\ell}+\bar{\sigma})\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)} \left\{\cos\left[\frac{1}{2}\left((\Delta\ell+\Delta\sigma)\phi_{q_{\parallel}}+\alpha\right)\right]\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) + \sin\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right] - i\sin\left[\frac{1}{2}\left((\Delta\ell+\Delta\sigma)\phi_{q_{\parallel}}+\alpha\right)\right]\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) - \sin\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]\right\}\hat{\boldsymbol{z}}$$

Define $\Delta J \equiv \Delta\ell + \Delta\sigma \quad \bar{J} \equiv \bar{\ell} + \bar{\sigma}$

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} e^{i\left(\bar{J}\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)} \left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_{\parallel}}+\alpha\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) + \tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right] - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_{\parallel}}+\alpha\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) - \tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]\right\}\hat{\boldsymbol{z}}$$

2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB:

$$\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{q}_{\parallel};\alpha,\beta) = -\frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\left((\bar{\ell}+\bar{\sigma})\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)} \left\{ \cos\left[\frac{1}{2}\left((\Delta\ell+\Delta\sigma)\phi_{q_{\parallel}}+\alpha\right)\right]\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) + \sin\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right] - i\sin\left[\frac{1}{2}\left((\Delta\ell+\Delta\sigma)\phi_{q_{\parallel}}+\alpha\right)\right]\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) - \sin\left(\frac{\beta}{2}\right)\tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]\right\}\hat{\boldsymbol{z}}$$

Define $\Delta J \equiv \Delta\ell + \Delta\sigma \quad \bar{J} \equiv \bar{\ell} + \bar{\sigma}$

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} e^{i\left(\bar{J}\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)} \left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_{\parallel}}+\alpha\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) + \tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right] - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_{\parallel}}+\alpha\right)\right]\left[\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) - \tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})\right]\right\}\hat{\boldsymbol{z}}$$

If $\tilde{F}_{\left|\bar{\ell}-\frac{\Delta\ell}{2}\right|}(q_{\parallel}) = \tilde{F}_{\left|\bar{\ell}+\frac{\Delta\ell}{2}\right|}(q_{\parallel})$

$$\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} \boxed{\tilde{F}_{\left|\bar{\ell}\pm\frac{\Delta\ell}{2}\right|}(q_{\parallel})}\, e^{i\left(\bar{J}\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)} \boxed{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_{\parallel}}+\alpha\right)\right]}\,\hat{\boldsymbol{z}}$$

Doughnut shape **Azimuthal structure**

2. Longitudinal component : Vector vortex beams (VVB)

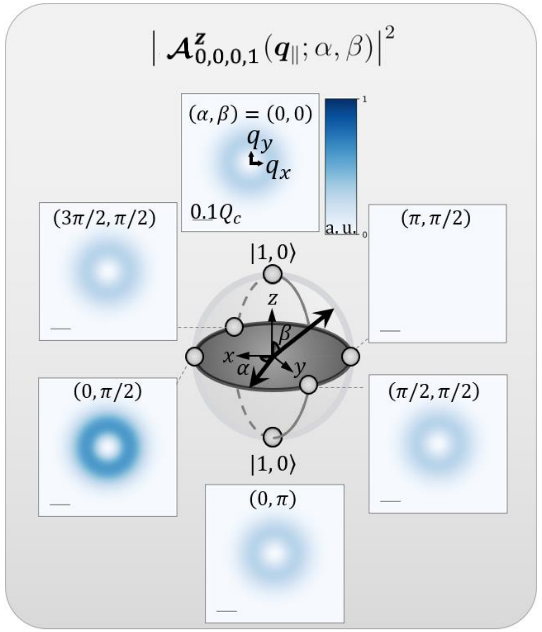
First case : $\ell = \ell'$

$$\mathcal{A}^{z}_{0,\ell,\Delta\sigma,\ell+\overline{\sigma}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2}\boxed{\tilde{F}_{|\ell|}(q_{\parallel})}\, e^{i\left((\ell+\overline{\sigma})\phi_{q_{\parallel}}+\frac{\alpha}{2}\right)}\boxed{\cos\left[\frac{1}{2}\left(\Delta\sigma\phi_{q_{\parallel}}+\alpha\right)\right]}\,\hat{\boldsymbol{z}}$$

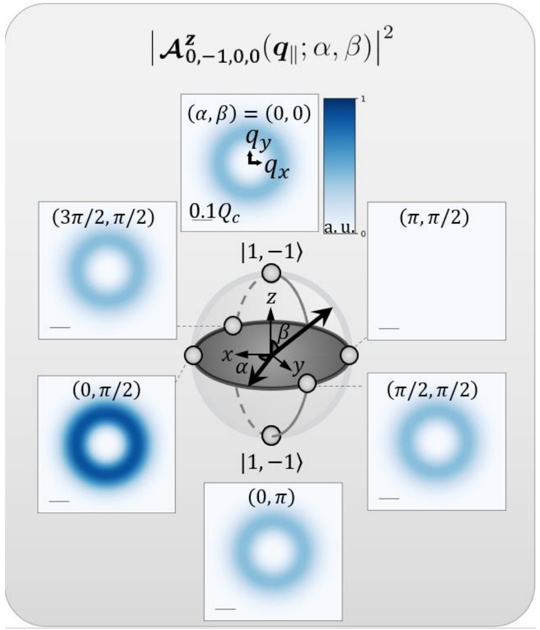
Doughnut shape **Azimuthal structure**

Same spin : $\Delta\sigma = 0$ $\Rightarrow$ **0 node**

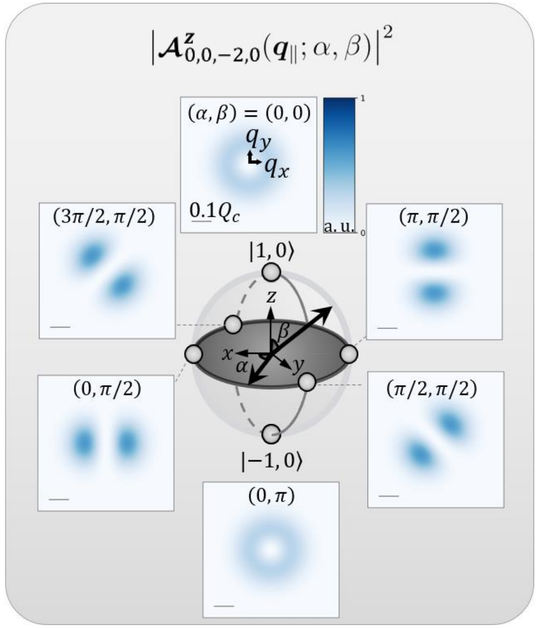
Opposite spin : $\Delta\sigma = -2$ $\Rightarrow$ **2 nodes**



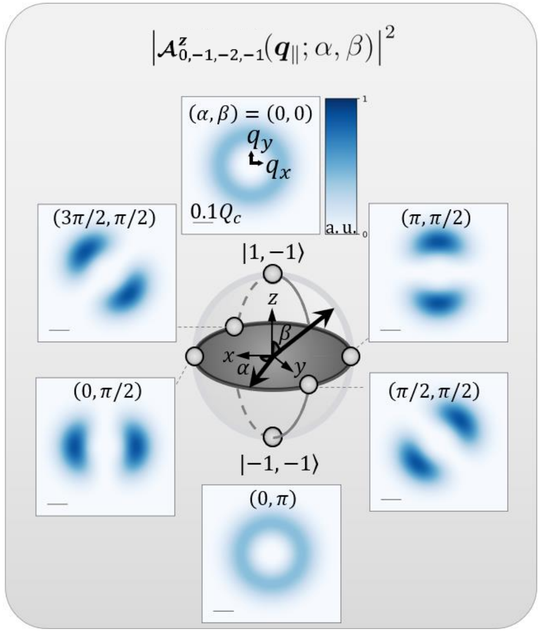
$\ell = 0$



$\ell = -1$



$\ell = 0$



$\ell = -1$

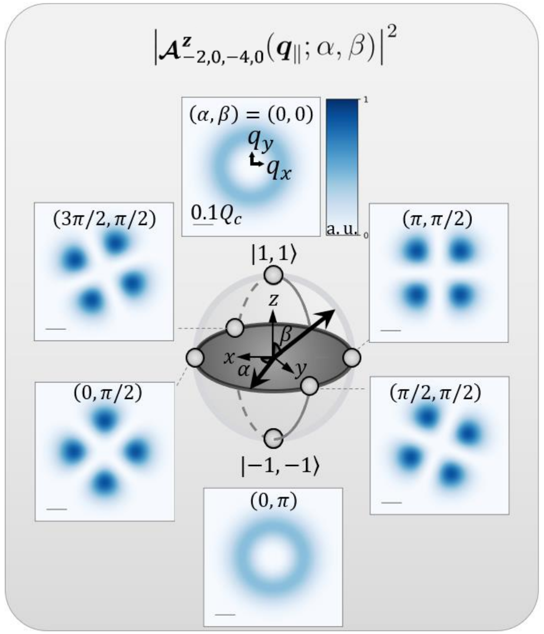
2. Longitudinal component : Vector vortex beams (VVB)

Second case : $\ell = -\ell'$

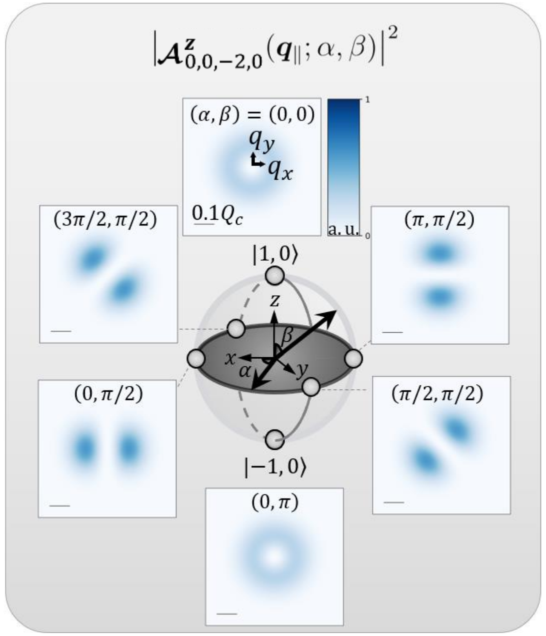
$$\boldsymbol{\mathcal{A}}^{\boldsymbol{z}}_{-2\ell,0,-2\ell+\Delta\sigma,\bar{\sigma}}\left(\boldsymbol{q}_\parallel;\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2}\boxed{\tilde{F}_{|\ell|}(q_\parallel)}\,e^{i\left(\bar{\sigma}\phi_{q_\parallel}+\frac{\alpha}{2}\right)}\boxed{\cos\left\{\frac{1}{2}\left[(-2\ell+\Delta\sigma)\phi_{q_\parallel}+\alpha\right]\right\}}\,\hat{\boldsymbol{z}}$$

Doughnut shape **Azimuthal structure**

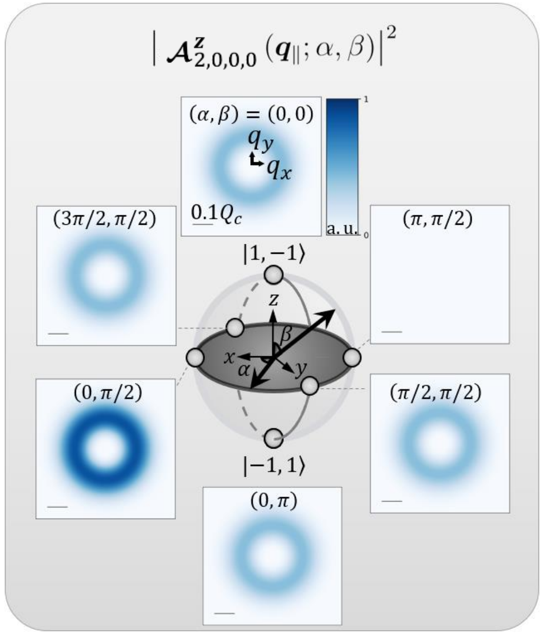
Total AM difference: $\Delta J = -2\ell + \Delta\sigma$ $\Rightarrow$ **even number of nodes**



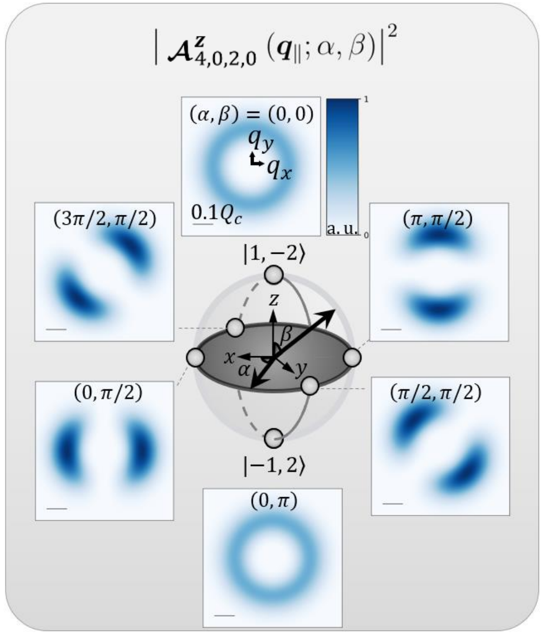
$\Delta J = -4$



$\Delta J = -2$



$\Delta J = 0$



$\Delta J = 2$

2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} e^{i\left(\bar{J}\phi_{q_\parallel}+\frac{\alpha}{2}\right)} \left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{|\ell|}(q_\parallel)+\tilde{F}_{|\ell'|}(q_\parallel)\right] - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{|\ell|}(q_\parallel)-\tilde{F}_{|\ell'|}(q_\parallel)\right]\right\}\hat{\boldsymbol{z}}$$

2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2}e^{i\left(\bar{J}\phi_{q_\parallel}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{|\ell|}(q_\parallel)+\tilde{F}_{|\ell'|}(q_\parallel)\right]-i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{|\ell|}(q_\parallel)-\tilde{F}_{|\ell'|}(q_\parallel)\right]\right\}\hat{\boldsymbol{z}}$$

Substitute $\tilde{F}_{|\ell|}(q_\parallel) = (-i)^{|\ell|}F_{|\ell|}(q_\parallel)$, and define $\overline{|\ell|} \equiv (|\ell'|+|\ell|)/2$, $\Delta|\ell| \equiv |\ell'|-|\ell|$

$$\mathcal{A}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2}e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]\left[F_{|\ell|}(q_\parallel)+F_{|\ell'|}(q_\parallel)\right]+i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]\left[F_{|\ell|}(q_\parallel)-F_{|\ell'|}(q_\parallel)\right]\right\}\hat{\boldsymbol{z}}$$

2. Longitudinal component : Vector vortex beams (VVB)

Can we have **odd** numbers of nodes ?

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{\mathbf{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} e^{i\left(\bar{J}\phi_{q_\parallel}+\frac{\alpha}{2}\right)} \left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{|\ell|}(q_\parallel)+\tilde{F}_{|\ell'|}(q_\parallel)\right] - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}+\alpha\right)\right]\left[\tilde{F}_{|\ell|}(q_\parallel)-\tilde{F}_{|\ell'|}(q_\parallel)\right]\right\}\hat{\mathbf{z}}$$

Substitute $\tilde{F}_{|\ell|}(q_\parallel) = (-i)^{|\ell|}F_{|\ell|}(q_\parallel)$, and define $\overline{|\ell|} \equiv (|\ell'|+|\ell|)/2$, $\Delta|\ell| \equiv |\ell'|-|\ell|$

$$\mathcal{A}^{\mathbf{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)} \left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]\left[F_{|\ell|}(q_\parallel)+F_{|\ell'|}(q_\parallel)\right] - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]\left[F_{|\ell|}(q_\parallel)-F_{|\ell'|}(q_\parallel)\right]\right\}\hat{\mathbf{z}}$$

Define $G_{|\ell|,|\ell'|}(q_\parallel) \equiv \left[F_{|\ell|}(q_\parallel)+F_{|\ell'|}(q_\parallel)\right]$ $\quad H_{|\ell|,|\ell'|}(q_\parallel) \equiv \left[F_{|\ell|}(q_\parallel)-F_{|\ell'|}(q_\parallel)\right]$

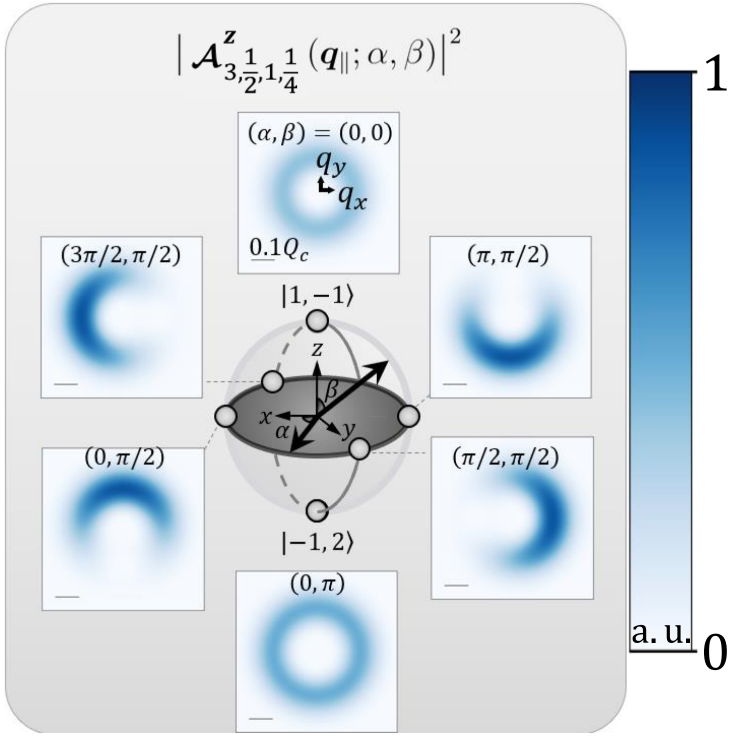
$$\mathcal{A}^{\mathbf{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)} \left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right] G_{|\ell|,|\ell'|}(q_\parallel) - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right] H_{|\ell|,|\ell'|}(q_\parallel)\right\}\hat{\mathbf{z}}$$

2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_\parallel;\alpha,\beta=\frac{\pi}{2}\right)=-\frac{\sin\theta_{\boldsymbol{q}}}{2}e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]G_{|\ell|,|\ell'|}(q_\parallel)-i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]H_{|\ell|,|\ell'|}(q_\parallel)\right\}\hat{\boldsymbol{z}}$$

Example for odd number ΔJ : $\sigma,\ell,\sigma',\ell' = 1,-1,-1,2$

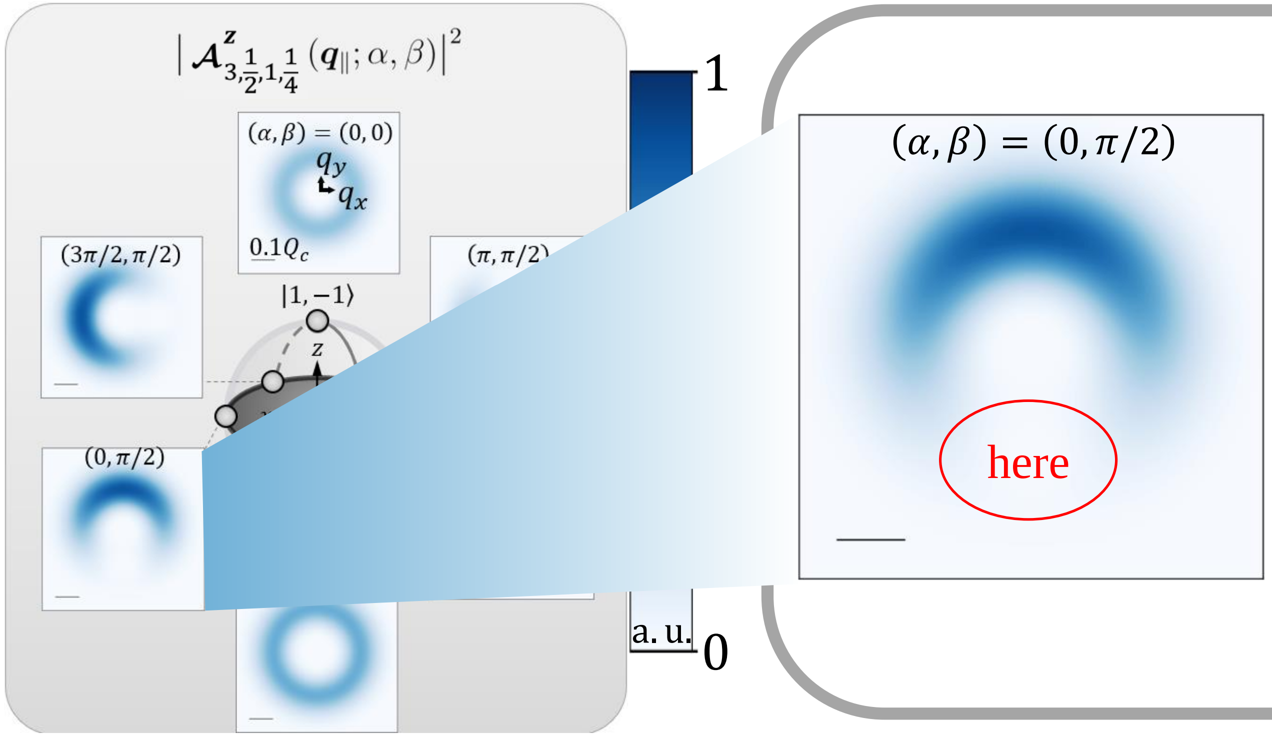


$\Delta J = 1$ **⇒ one node**

2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_\parallel;\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2} e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)} \left\{ \boxed{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]} G_{|\ell|,|\ell'|}(q_\parallel) - \boxed{i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]} H_{|\ell|,|\ell'|}(q_\parallel) \right\}\hat{\boldsymbol{z}}$$



$\sigma,\ell,\sigma',\ell' = 1,-1,-1,2$

But ! For $\alpha = 0$, node appear when $\phi_{q_\parallel} = \frac{3\pi}{2}$

$$\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right] = \cos\left[\frac{\pi}{2}\right] = 0$$

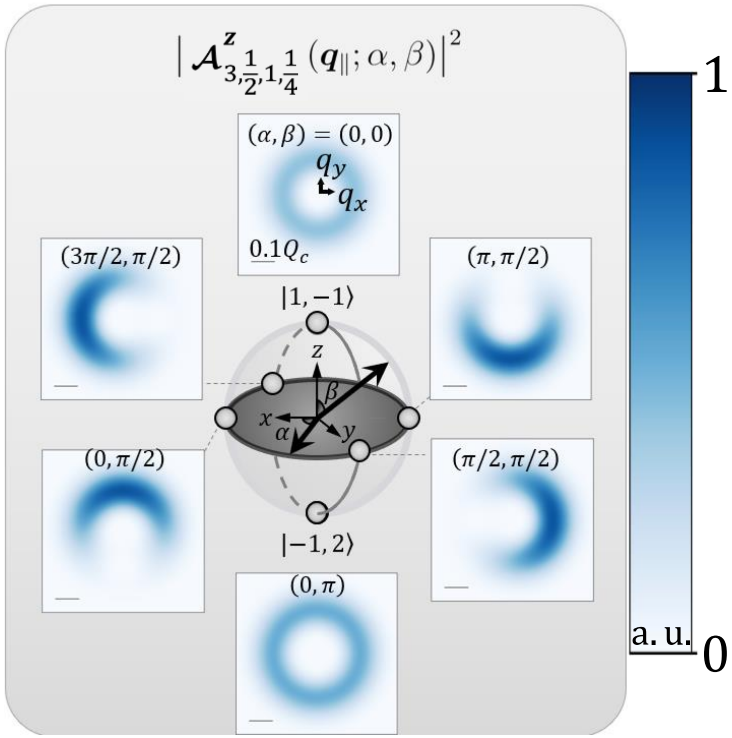
$$\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right] = \sin\left[\frac{\pi}{2}\right] = \mathbf{1}$$

Maximum value !?

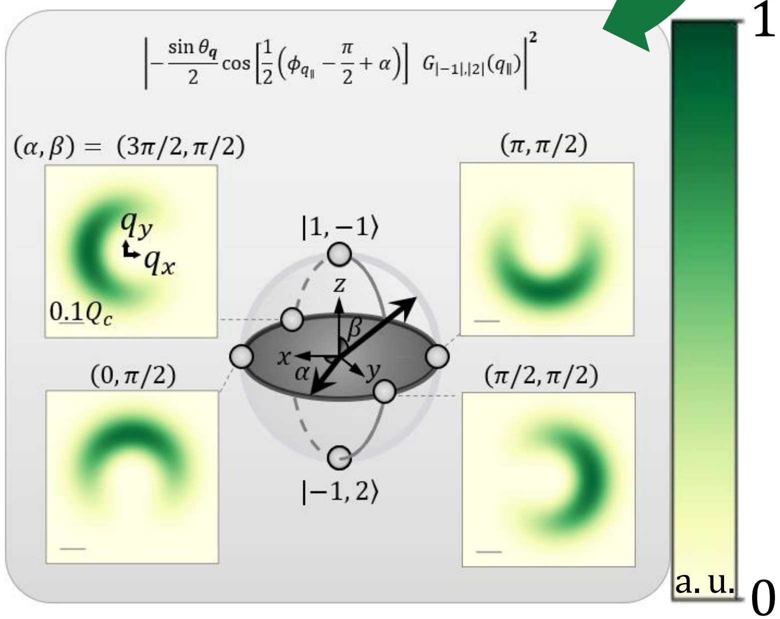
2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

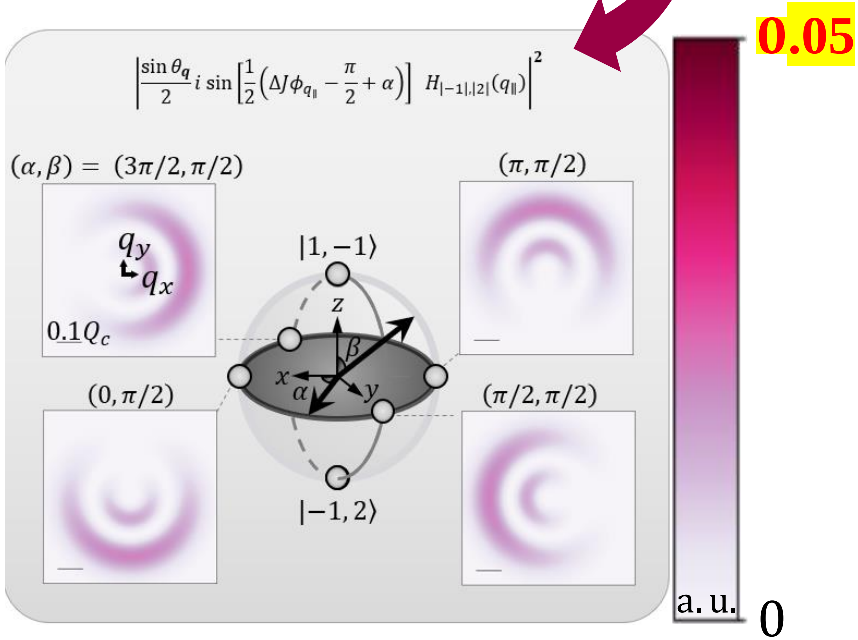
$$\mathcal{A}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_\parallel;\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2}e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)}\left\{\boxed{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]\ G_{|\ell|,|\ell'|}(q_\parallel)}-\boxed{i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]\ H_{|\ell|,|\ell'|}(q_\parallel)}\right\}\hat{\boldsymbol{z}}$$



$\sigma,\ell,\sigma',\ell' = 1,-1,-1,2$



$G_{|\ell|,|\ell'|}(q_\parallel) \equiv \left[F_{|\ell|}(q_\parallel) + F_{|\ell'|}(q_\parallel)\right]$



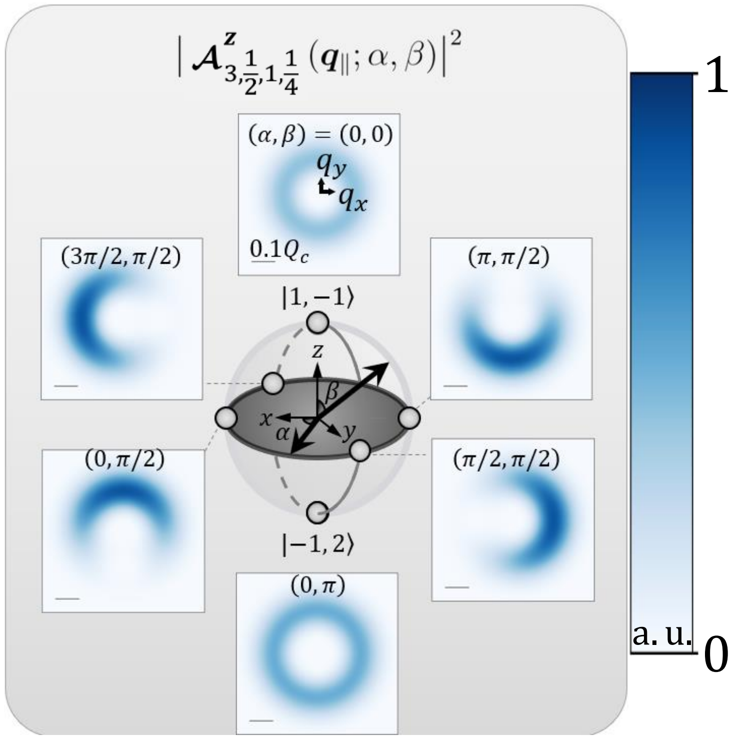
$H_{|\ell|,|\ell'|}(q_\parallel) \equiv \left[F_{|\ell|}(q_\parallel) - F_{|\ell'|}(q_\parallel)\right]$

Nearly zero

2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{\boldsymbol{z}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2}\, e^{i\left(\bar{J}\phi_{q_{\parallel}}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_{\parallel}}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right] G_{|\ell|,|\ell'|}(q_{\parallel}) - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_{\parallel}}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right] H_{|\ell|,|\ell'|}(q_{\parallel})\right\}\hat{\boldsymbol{z}}$$



$\sigma,\ell,\sigma',\ell' = 1,-1,-1,2$

What does it means to have a horseshoe shape light?

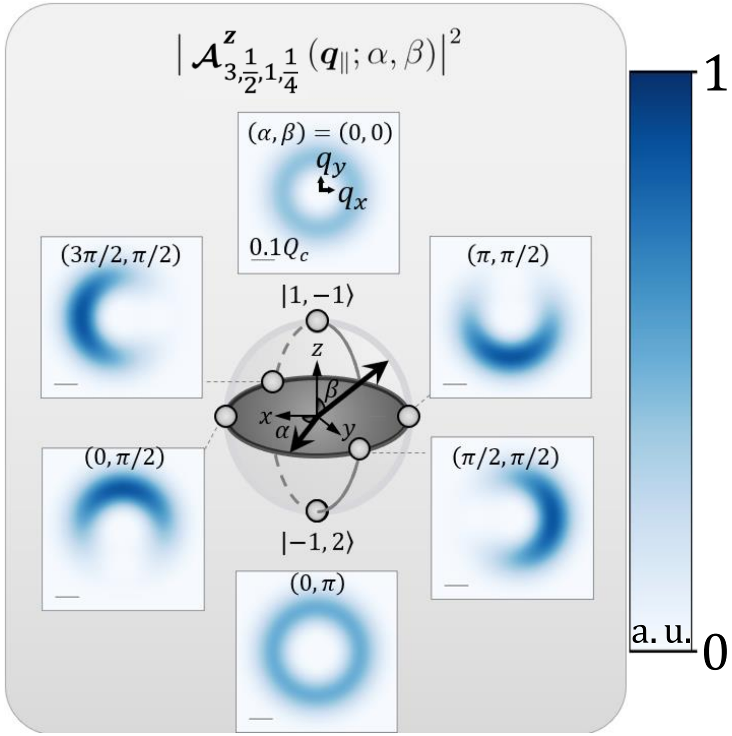
2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

$$\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right)=-\frac{\sin\theta_{\boldsymbol{q}}}{2}e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]G_{|\ell|,|\ell'|}(q_\parallel)-i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]H_{|\ell|,|\ell'|}(q_\parallel)\right\}\hat{\boldsymbol{z}}$$

Find the real field !

$$\mathrm{Re}\left[\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right)e^{-i\omega t}\right]$$

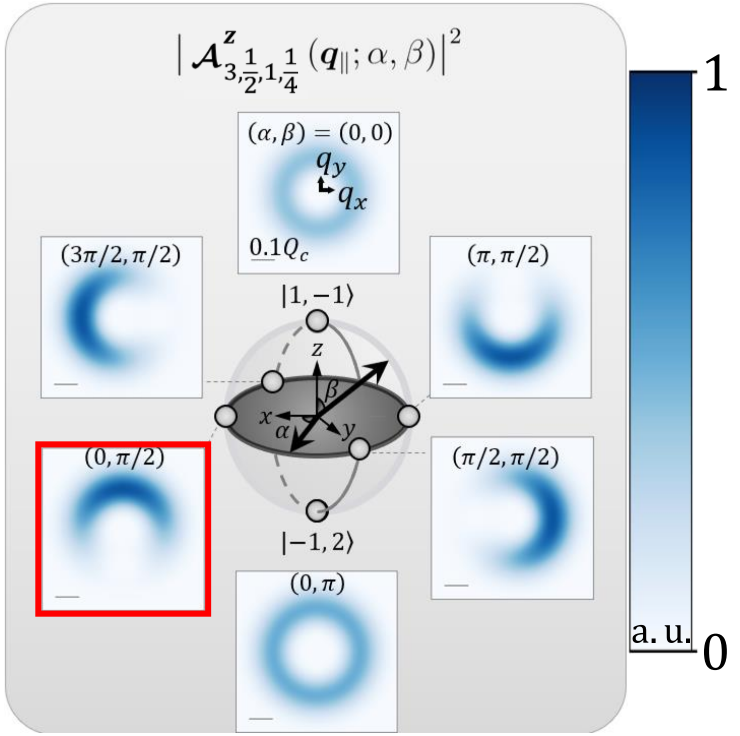


$\sigma,\ell,\sigma',\ell' = 1,-1,-1,2$

2. Longitudinal component : Vector vortex beams (VVB)

Longitudinal component of VVB: $\beta = \frac{\pi}{2}$

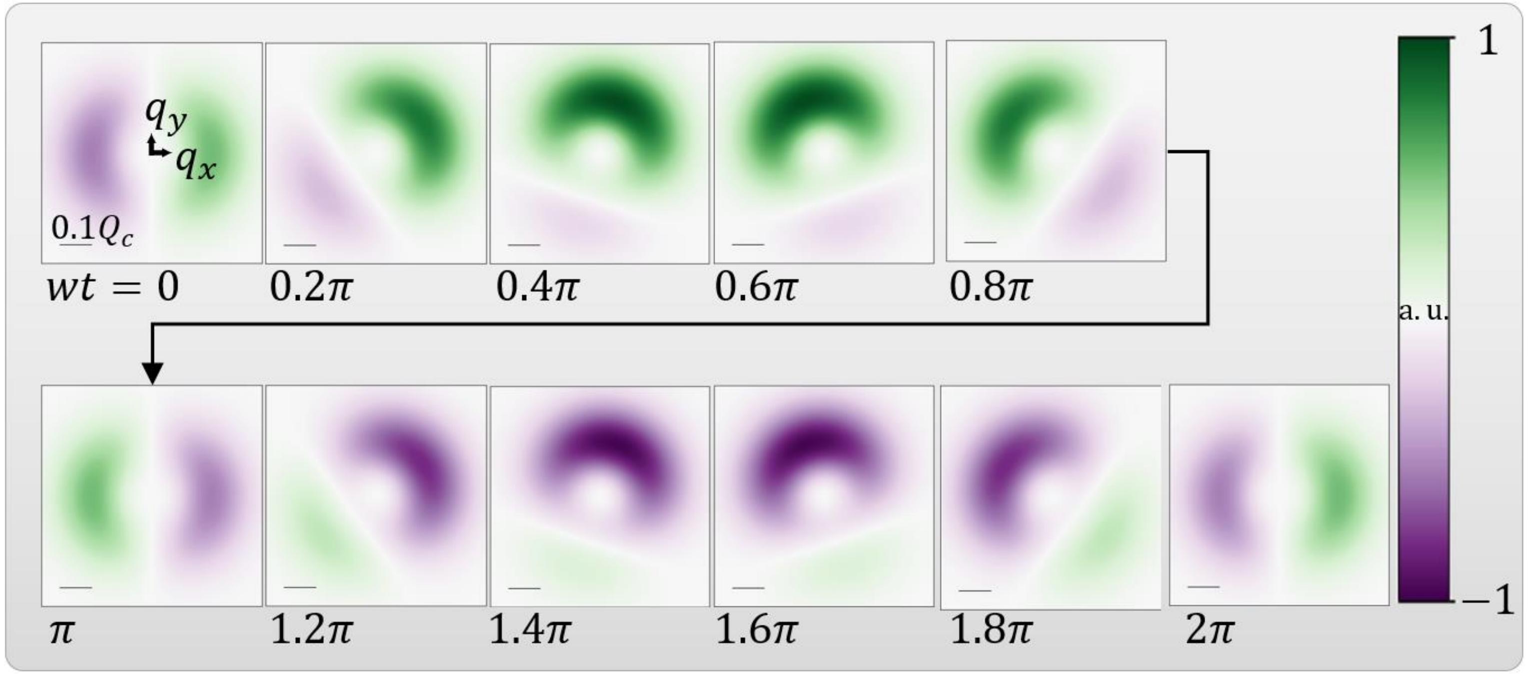
$$\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right) = -\frac{\sin\theta_{\boldsymbol{q}}}{2}e^{i\left(\bar{J}\phi_{q_\parallel}-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)}\left\{\cos\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]G_{|\ell|,|\ell'|}(q_\parallel) - i\sin\left[\frac{1}{2}\left(\Delta J\phi_{q_\parallel}-\frac{\pi}{2}\Delta|\ell|+\alpha\right)\right]H_{|\ell|,|\ell'|}(q_\parallel)\right\}\hat{\boldsymbol{z}}$$



$\sigma,\ell,\sigma',\ell' = 1,-1,-1,2$

Find the real field !

$$\mathrm{Re}\left[\mathcal{A}^{z}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}\left(\boldsymbol{q}_{\parallel};\alpha,\beta=\frac{\pi}{2}\right)e^{-i\omega t}\right]$$



Outline

3. Light-matter interaction

Outline

4. Exciton wave packet

Summary and outlook

- Summary and outlook